

Physiologically-Informed Time-Block Planning

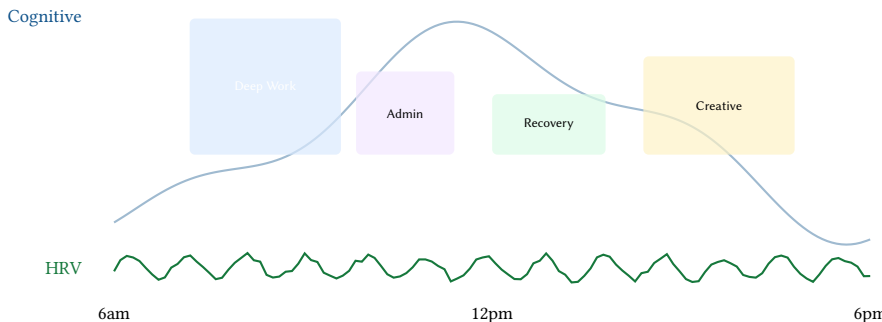
A Smart Day Planner Integrating Circadian Rhythms, Recovery State, and Task Semantics

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Abstract. Traditional task management systems treat all hours as interchangeable, ignoring the well-established circadian variation in cognitive performance, the physiological cost of physical activity, and the compounding effects of sleep debt. This report presents *WeekPlanner*, a system that synthesizes task scheduling from three signal domains: (1) a circadian cognitive model parameterised by individual chronotype, (2) real-time physiological state derived from heart-rate variability and sleep metrics, and (3) semantic task metadata from OmniFocus including effort, context, and deadline constraints. Drawing on Cal Newport’s time-block planning methodology, the system produces daily and weekly schedules that align deep-work sessions with peak cognitive windows, enforce pre-activity tapering and post-activity recovery, and respect the finite nature of daily willpower. The architecture reuses components from an existing agent ecosystem including health signal processing, OmniFocus database access, and LLM-based task orchestration. We present the formal scheduling model, the constraint system, the recovery science underpinning activity pacing, and the planned architecture with detailed data flows.

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WP	WeekPlanner	2
HRV	Heart Rate Variability	2
RMSSD	Root Mean Square of Successive Differences	2
EWMA	Exponentially Weighted Moving Average	4
LLM	Large Language Model	8
GTD	Getting Things Done	2
SCN	Suprachiasmatic Nucleus	2
TST	Total Sleep Time	4
DAG	Directed Acyclic Graph	39
CSP	Constraint Satisfaction Problem	3
ANS	Autonomic Nervous System	6
DOMS	Delayed Onset Muscle Soreness	6
PNS	Parasympathetic Nervous System	6
SNS	Sympathetic Nervous System	6

1 Motivation: Why Time Is Not Fungible

Task management methodologies from Getting Things Done (GTD) [1] to Cal Newport’s time-block planning [2] share a common insight: *what* you work on matters less than *when* you work on it. Yet most digital planners—OmniFocus [4], Todoist, Things—model tasks as context-tagged items in a priority queue, with no awareness of the human executing them.

The result is a planning gap analogous to the “Verification Gap” in network synthesis: intent (the task list) is decoupled from execution reality (the human’s cognitive and physiological state). WeekPlanner (WP) closes this gap by treating the day not as a blank canvas but as a *typed resource* with measurable capacity that varies hour by hour.

Insight:
A 90-minute deep-work block at 10 am produces more than three scattered hours in the afternoon [3].

1.1 The Conventional Task Pipeline

To understand the gap, consider how existing tools process tasks. A conventional pipeline moves items from capture through prioritisation to execution, but the execution context is entirely absent from the model.

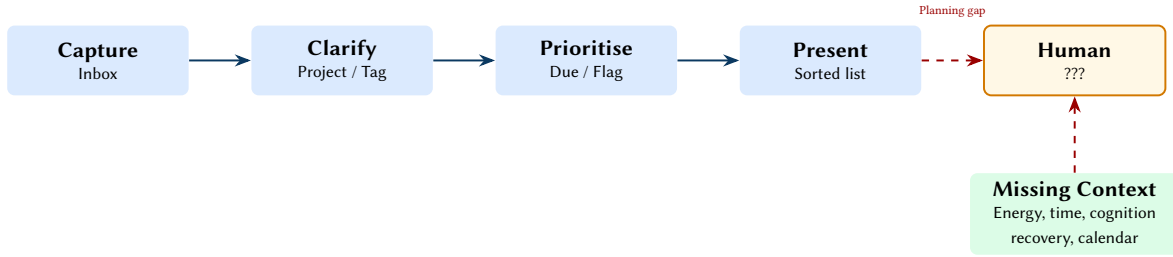


Figure 1. The conventional task pipeline. Tasks flow from capture to a prioritised list, but the final step—mapping to the human’s actual day—has no system support. The planning gap is where WP operates.

1.2 Evidence Base

Three bodies of evidence motivate the WP approach:

1. **Circadian cognition.** Alertness, working memory, and executive function follow a predictable diurnal curve governed by the Suprachiasmatic Nucleus (SCN) [5]. For most chronotypes, peak cognitive performance occurs 2–4 hours after waking, with a well-documented post-lunch dip [6] and a secondary peak in the late afternoon.
2. **Physiological recovery.** Heart Rate Variability (HRV) metrics—particularly Root Mean Square of Successive Differences (RMSSD)—are reliable proxies for parasympathetic tone and recovery state [7]. A morning HRV reading below personal baseline signals accumulated fatigue, suggesting the day’s plan should shift toward lighter cognitive load.
3. **Willpower depletion.** The strength model of self-control posits that effortful cognitive tasks draw from a shared, finite resource [8]. Scheduling decisions should front-load demanding work and reserve routine tasks for depleted periods.

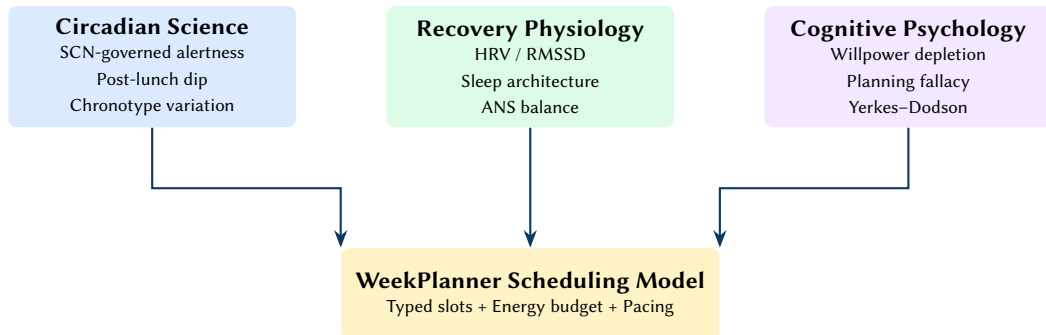


Figure 2. Three evidence pillars converge into the WP scheduling model. Each pillar contributes distinct constraints: circadian science types time slots, recovery physiology modulates daily budgets, and cognitive psychology shapes task ordering and pacing.

: Temporal Typing

Every hour-slot h_i in the daily schedule carries a cognitive-capacity type $\kappa(h_i) \in \{\text{Peak, Sustain, Trough, Recovery}\}$ derived from the circadian model. A task requiring deep focus *must not* be assigned to a Trough-typed slot.

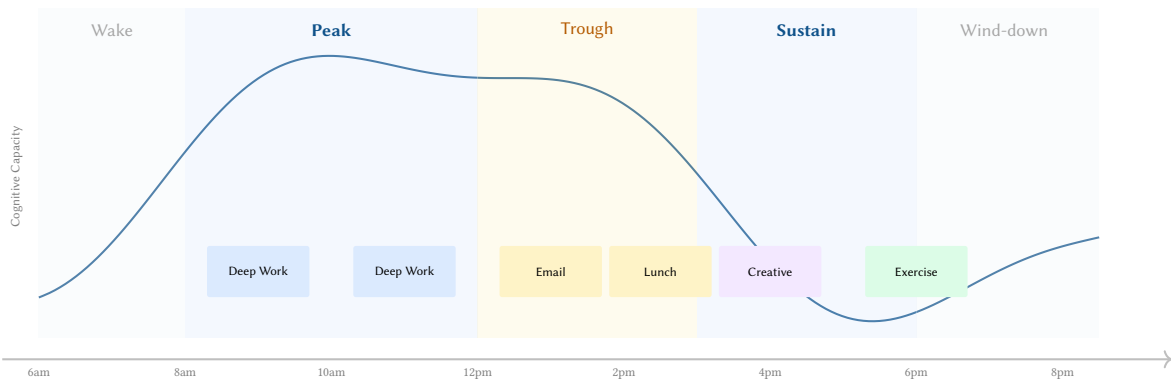


Figure 3. Circadian capacity zones mapped to time-block types. The sinusoidal curve represents the aggregate cognitive capacity model; coloured bands indicate the derived slot types. Tasks are assigned to compatible zones.

2 Related Approaches

Several methodologies inform the WP design.

2.1 Cal Newport’s Time-Block Planning

Newport advocates allocating every minute of the working day to a specific activity block [2]. The method is paper-based: a ruled notebook with time on the left axis and block labels drawn as rectangles. Key principles include:

- **Deep work batching.** Group cognitively demanding tasks into uninterrupted 90-minute blocks, ideally in the morning.
- **Shallow-work containment.** Batch email, Slack, and administrative tasks into designated blocks to prevent context switching.
- **Shutdown complete.** End each workday with a ritual that captures incomplete items and sets the next morning’s plan, enabling psychological detachment [9].

2.2 Getting Things Done (GTD)

Allen’s GTD framework [1] provides the task *capture* and *organisation* layers that WP builds upon. OmniFocus is a faithful GTD implementation, providing projects, contexts, defer/due dates, and sequential/parallel action groups. WP does not replace GTD—it consumes its output.

Insight:
WP digitises Newport’s paper method while adding the physiological dimension he acknowledges but cannot capture on paper.

2.3 Comparison Matrix

	Task Capture	Time Slots	Energy	Recovery	Weekly Pacing	Adaptive
GTD	✓	—	—	—	—	—
Time-Block	manual	✓	—	—	—	—
WeekPlanner	✓	✓	✓	✓	✓	✓

Figure 4. Feature comparison across planning methodologies. WP consumes GTD’s task capture and extends time-block planning with physiological awareness, recovery modelling, and adaptive replanning.

3 The Scheduling Model

WP models the scheduling problem as a Constraint Satisfaction Problem (CSP) over typed time slots.

Insight:
By framing scheduling as constraint satisfaction rather than optimisation, we avoid brittle “optimal” plans that shatter on first contact with reality.

3.1 Formal Definitions

Let $\mathcal{D}$ denote a day partitioned into n slots of duration δ (default 30 min):

$$\mathcal{D} = \{h_1, h_2, \dots, h_n\}$$

Each slot carries a capacity type derived from the circadian model $\mathcal{C}$ and modulated by the physiological state Φ :

$$\kappa(h_i) = \mathcal{C}(t_i, \theta) \otimes \Phi_{\text{morning}}$$

where θ is the individual's chronotype parameter and Φ_{morning} is the day's recovery state from the morning HRV reading.

A task τ_j from the task pool $\mathcal{T}$ carries:

- **Effort type** $\epsilon(\tau_j) \in \{\text{Deep, Shallow, Physical, Recovery}\}$
- **Duration estimate** $d(\tau_j)$ in slot units
- **Deadline** $\delta(\tau_j)$ (optional, from OmniFocus)
- **Context tags** $\gamma(\tau_j)$ (location, tools required)
- **Energy cost** $e(\tau_j) \in [0, 1]$ (normalised drain)

Axiom 1 (Capacity Compatibility). A task τ_j may be assigned to slot h_i only if $\epsilon(\tau_j) \preceq \kappa(h_i)$, where $\preceq$ is the compatibility partial order: $\text{Deep} \preceq \text{Peak}$, $\text{Shallow} \preceq \{\text{Peak, Sustain, Trough}\}$, $\text{Physical} \preceq \{\text{Peak, Sustain}\}$, $\text{Recovery} \preceq \{\text{any}\}$.

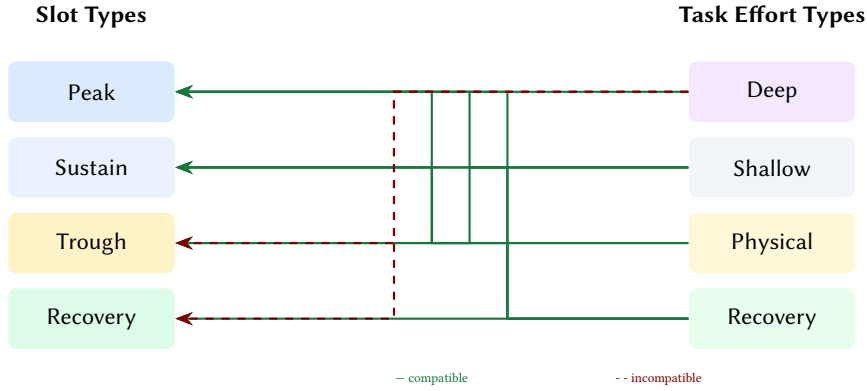


Figure 5. Capacity compatibility partial order. Green solid lines indicate permitted assignments; red dashed lines indicate violations. Deep tasks require Peak slots; Recovery tasks fit anywhere.

3.2 Energy Budget

The day's total energy budget E_{total} is derived from sleep and recovery signals:

$$E_{\text{total}} = E_{\text{base}} \cdot \alpha(\text{HRV}_{\text{morning}}) \cdot \beta(\text{Total Sleep Time (TST)})$$

where α maps HRV to a $[0.6, 1.2]$ multiplier via the personal Exponentially Weighted Moving Average (EWMA) baseline [7], and β penalises sleep debt using the two-process model [10].

: Energy Conservation

The sum of energy costs for all scheduled tasks must not exceed the daily budget: $\sum_j e(\tau_j) \leq E_{\text{total}}$. When budget is exhausted, remaining slots are typed as Recovery.

3.3 Willpower Drain Model

Beyond the per-task energy cost, the system models cumulative willpower drain across the day. After each deep-work block, the effective capacity of subsequent Peak slots degrades:

$$\kappa_{\text{eff}}(h_i) = \kappa(h_i) \cdot \left(1 - \lambda \sum_{j < i} e(\tau_j) \cdot [\epsilon(\tau_j) = \text{Deep}]\right)$$

where λ is the willpower-drain coefficient (default 0.15 per unit of deep-work energy spent).

Insight:
This prevents the naive strategy of scheduling 8 consecutive hours of deep work—by hour 6, effective capacity has degraded below the Deep threshold.

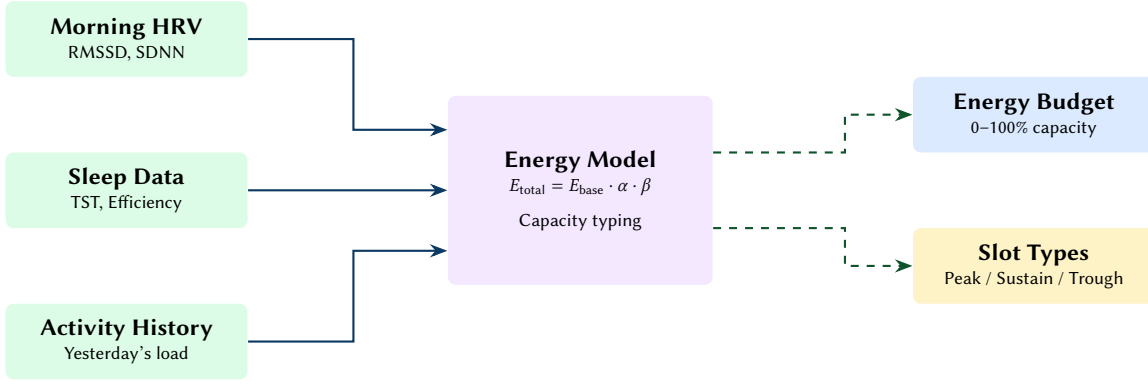


Figure 6. The Energy Model derives daily capacity and slot types from three physiological signal sources. Orthogonal routing reflects the clean separation of input signals and output channels.

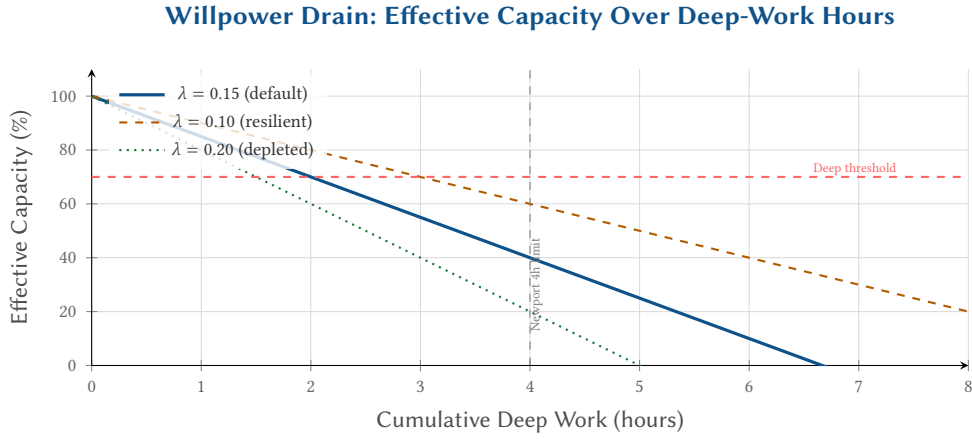


Figure 7. Willpower drain curves for three profiles. The horizontal dashed line marks the threshold below which Deep tasks should not be scheduled. Newport’s empirical 4-hour limit aligns with $\lambda = 0.15$.

4 Circadian Model

The circadian cognitive model is parameterised by chronotype $\theta \in [-2, 2]$, where $\theta = 0$ is the population-average (“intermediate”) type, negative values represent early chronotypes (“larks”), and positive values represent late chronotypes (“owls”).

The base cognitive capacity at clock time t is:

$$C_{\text{base}}(t, \theta) = A \sin(\omega(t - \phi_0 - \theta)) + B \sin(2\omega(t - \phi_1)) + C_0$$

where A and B weight the primary and post-lunch-dip harmonics, ϕ_0 and ϕ_1 are phase offsets, $\omega = 2\pi/24$, and C_0 is the baseline capacity [3, 5].

4.1 Physiological Modulation

The base model is adjusted each morning by actual recovery state:

$$C_{\text{actual}}(t) = C_{\text{base}}(t, \theta) \cdot \alpha(\text{HRV}) \cdot \beta(\text{TST})$$

Insight:
The two-harmonic model captures both the main circadian peak and the post-lunch dip, which single-harmonic models miss.

Table 1. Recovery modulation factors.

Signal	Good Day	Depleted Day
$\alpha(\text{HRV})$: RMSSD vs 30-day EWMA	≥ 1.0	0.6–0.8
$\beta(\text{TST})$: Sleep vs 7.5 h target	1.0	0.85^k (k = debt nights)

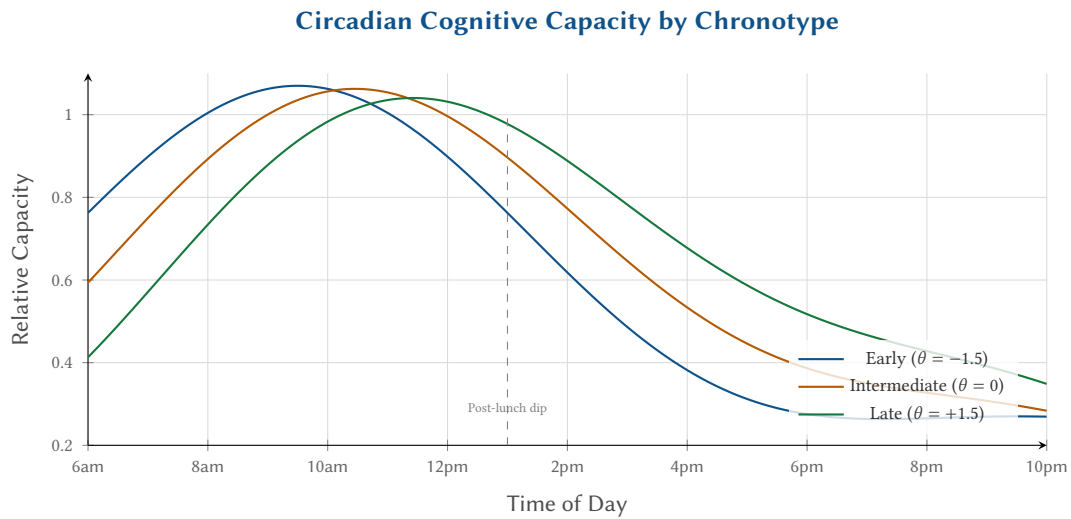


Figure 8. Two-harmonic circadian model for three chronotypes. Early types peak around 9–10 am; late types peak around 11 am–12 pm. All exhibit a post-lunch dip centred near 1–2 pm.

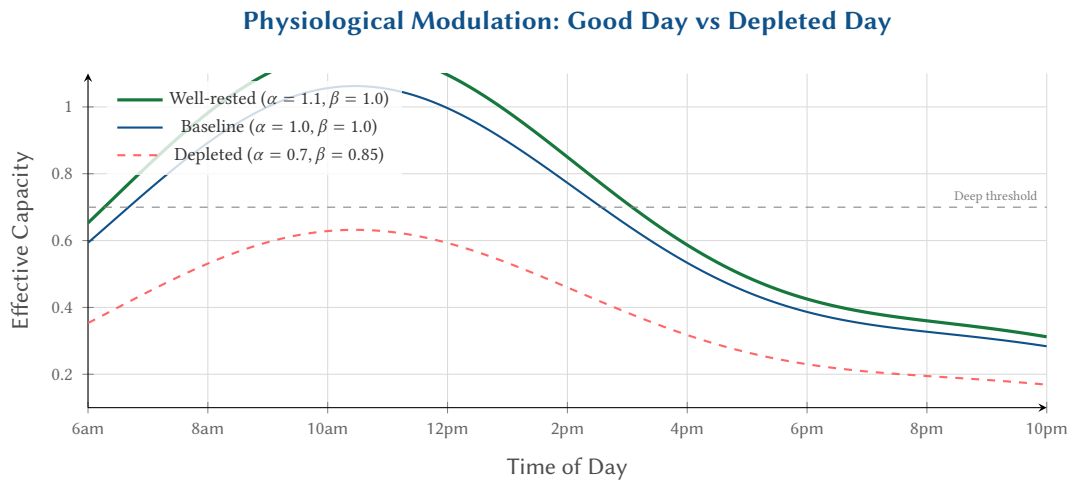


Figure 9. Physiological modulation compresses or expands the circadian curve. On a depleted day, the effective Peak window shrinks significantly, and the system responds by reducing deep-work allocation.

5 Recovery Science and Activity Pacing

A distinguishing feature of **WP** is its explicit modelling of physical recovery. Most planners treat exercise as “just another task”; **WP** treats it as a physiological event with multi-day consequences.

5.1 The Autonomic Stress–Recovery Cycle

Physical exertion shifts the Autonomic Nervous System (**ANS**) toward Sympathetic Nervous System (**SNS**) dominance (fight-or-flight). Recovery requires a return to Parasympathetic Nervous System (**PNS**) dominance (rest-and-digest), measurable via **HRV** [7]. The timeline depends on intensity:

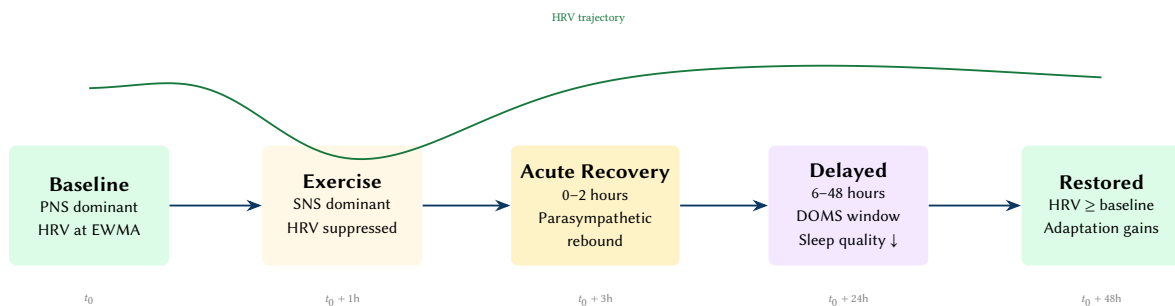


Figure 10. The autonomic stress–recovery cycle following high-intensity exercise. **HRV** suppression during exercise is followed by acute rebound, then a delayed recovery window where Delayed Onset Muscle Soreness (**DOMS**) and sleep disruption may occur before full restoration.

5.2 Activity Classification

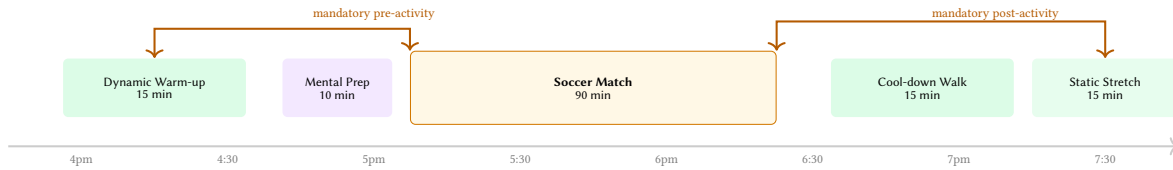
WP classifies physical activities by recovery cost:

Table 2. Activity classification and recovery parameters.

Activity	Intensity	Recovery (h)	Taper?	Next-Day ρ
Soccer match	High	36–48	Yes (24h)	0.65
Gym (strength)	High	24–36	No	0.75
Running (> 5km)	Moderate+	18–24	Optional	0.80
Cycling (indoor)	Moderate	12–18	No	0.85
Yoga / stretching	Low	0	No	1.05
Walking (< 30min)	Minimal	0	No	1.00

5.3 Stretching and Mobility Windows

The system inserts mandatory mobility blocks around high-intensity activities:



Insight:
Yoga and gentle walking are net-positive for recovery—they actively restore PNS tone and improve next-day capacity, reflected in $\rho > 1.0$.

Figure 11. Mandatory mobility windows around a high-intensity event. The system automatically inserts warm-up and cool-down blocks and prevents scheduling cognitive work in adjacent slots.

6 Weekly Planning Constraints

The weekly view introduces inter-day constraints that the daily planner must respect.

6.1 Activity Tapering and Recovery

Physical activities impose constraints on adjacent days:

: Pre-Activity Taper

If day d_k contains a high-intensity physical event (e.g. soccer match), then d_{k-1} must cap physical energy expenditure at 40% of baseline and include a minimum 30-minute recovery block (stretching, mobility).

: Post-Activity Recovery

Day d_{k+1} after a high-intensity event receives a recovery modifier $\rho = 0.7$, reducing both cognitive and physical budgets. A dedicated recovery block (yoga, walking) is mandatory.

Insight:
Weekly constraints prevent the common failure mode of treating each day as independent, leading to back-to-back high-intensity days.

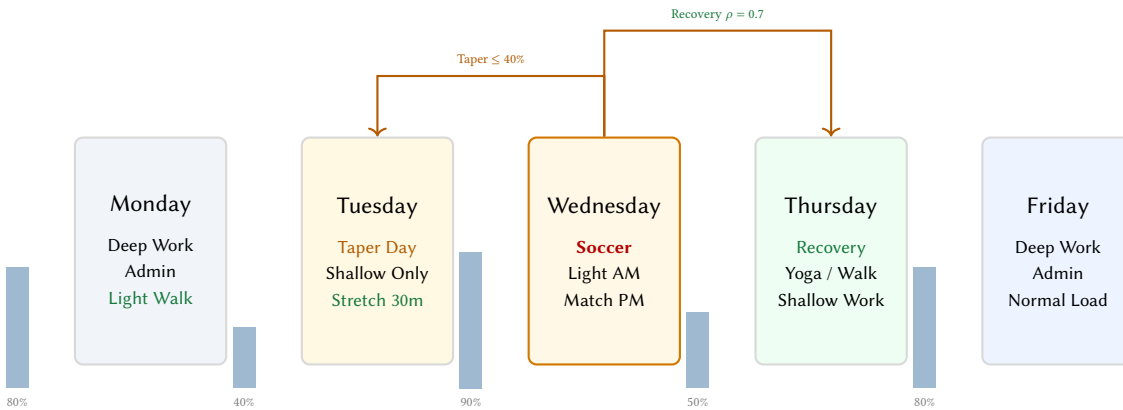


Figure 12. Weekly constraint propagation around a high-intensity event (soccer). Orthogonal arrows show taper and recovery constraints propagating from the event day to adjacent days. Energy bars (left of each day) show the resulting budget allocation.

6.2 Pacing Rules

Beyond single-event recovery, WP enforces global pacing:

- **No consecutive peak days.** At most 2 consecutive days at > 85% energy expenditure; day 3 is automatically downgraded.
- **Weekly deep-work cap.** Following Newport’s recommendation of ≈ 4 hours/day of deep work for experienced practitioners [2], the weekly deep-work budget is capped at 20 hours (adjustable).
- **Recovery ratio.** For every 3 hours of high-intensity (physical or cognitive), at least 1 hour of explicit recovery must appear within 48 hours.
- **Sleep-debt accumulation.** If TST falls below 7 hours for 2+ consecutive nights, all next-day budgets receive a cumulative 0.85^k penalty (where k = consecutive short nights).

Pacing Invariant

For any rolling 7-day window W :

$$\frac{\sum_{d \in W} E_{\text{spent}}(d)}{\sum_{d \in W} E_{\text{total}}(d)} \leq 0.80$$

The system maintains a 20% energy reserve to absorb unexpected demands.

Weekly Energy Budget Allocation (Example Week)

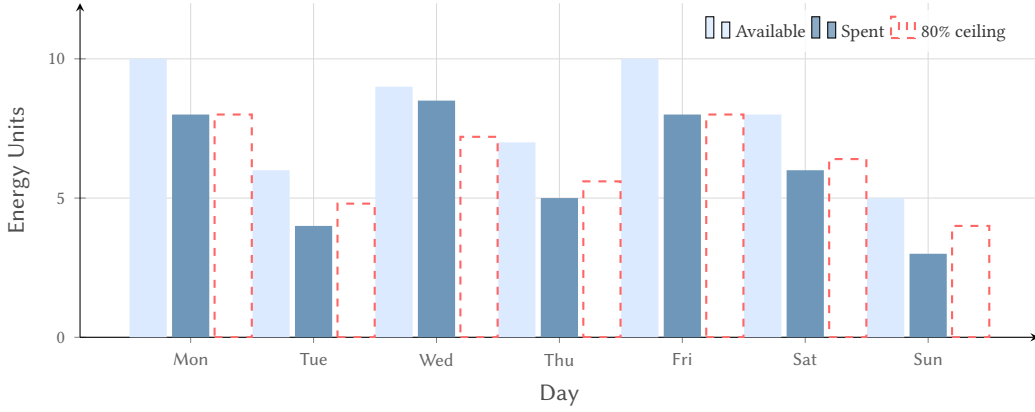


Figure 13. Weekly energy budget and expenditure. Tuesday and Thursday show reduced budgets due to taper/recovery around Wednesday’s soccer match. Sunday is deliberately light for weekly review and restoration.

7 Task Classification

Tasks imported from OmniFocus arrive with project, context tags, and due dates but no cognitive-load metadata. WP enriches them via a two-stage classifier.

7.1 Rule-Based Pre-Classification

Tag and project patterns map to effort types:

Table 3. Rule-based task classification heuristics.

Pattern	Effort Type	Energy Cost
Tag: “deep”, “writing”, “coding”	Deep	0.7–0.9
Tag: “email”, “admin”, “errands”	Shallow	0.1–0.3
Tag: “exercise”, “soccer”, “gym”	Physical	0.6–0.9
Tag: “yoga”, “stretch”, “walk”	Recovery	0.0–0.1
Project: “PhD”, “research”	Deep	0.8
Duration > 90 min	Deep	scaled

7.2 LLM Refinement

Tasks not matched by rules are classified by the Large Language Model (LLM) planner, which sees the task title, notes, and project context.

Insight:
The LLM acts as a fallback classifier, not the primary one—deterministic rules handle $\approx 80\%$ of tasks, keeping the system predictable.

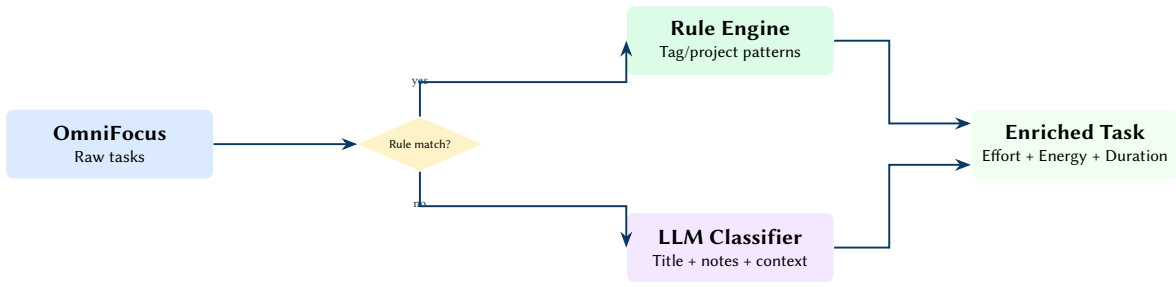
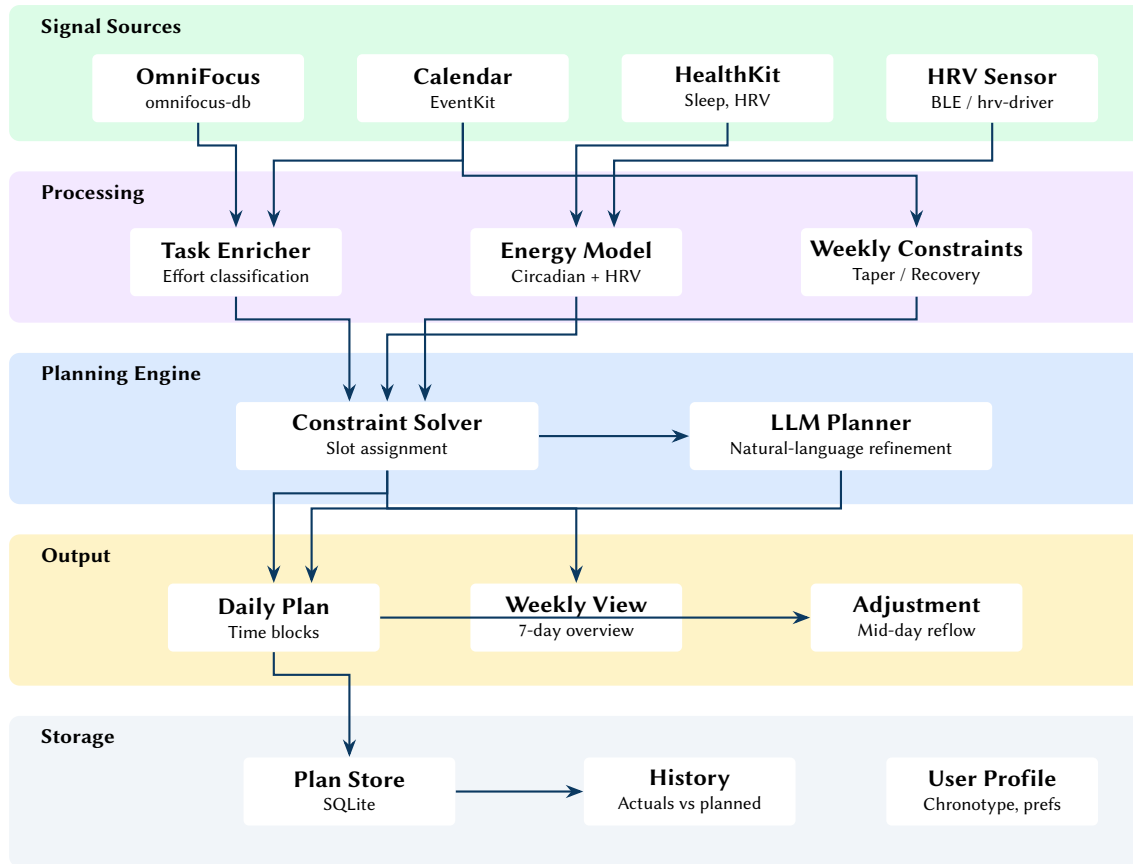


Figure 14. Two-stage task classification pipeline. Rule-based matching handles the majority of tasks deterministically; unmatched tasks fall through to the LLM classifier for semantic analysis.

8 System Architecture

WP is composed from existing components in the personal-tools ecosystem, connected by a lightweight orchestration layer.



Insight:
The architecture reuses proven subsystems rather than rebuilding—the OmniFocus reader, health signal pipeline, and LLM planner all exist as standalone tools.

Figure 15. WeekPlanner five-layer system architecture with orthogonal signal routing. Existing ecosystem components (omnifocus-db, hrv-driver, healthypi) provide the signal source and processing layers.

8.1 Component Mapping to Existing Projects

Table 4. Reuse mapping: existing ecosystem components to WeekPlanner roles.

WP Component	Source Project	Language	Reuse Strategy
Task source	omnifocus-db	Python	Direct import
Task model	todo (gsd)	Rust	Domain types via MCP
Task triage	omni	Python	AI matching patterns
HRV signals	hrv-driver	Rust	BLE capture, Parquet
Health analysis	healthypi	Python	HRV/sleep/stress modules
Energy baselines	healthypi	Python	EWMA baseline engine
LLM planning	multi-agent	Python	Tool-calling patterns
Messaging	multi-agent	Python	NATS (optional)

8.2 Data Model

The core data model centres on three entities: *Slot*, *Task*, and *Plan*. Their relationships enforce the scheduling invariants at the data level.

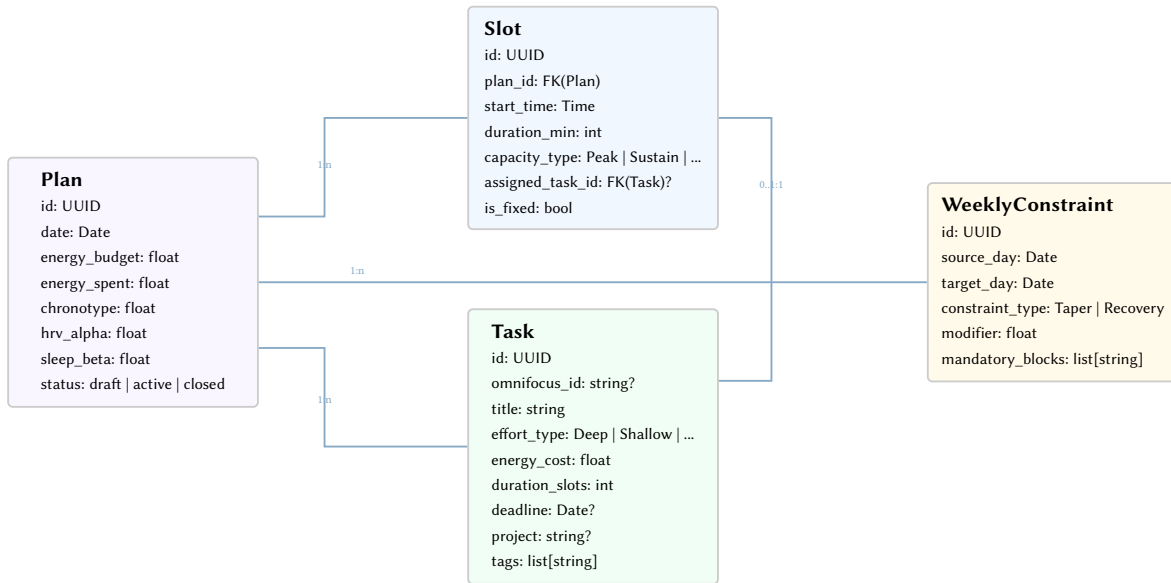


Figure 16. Core data model. A Plan contains Slots and references Tasks. WeeklyConstraints propagate modifiers between days. The assigned_task_id on Slot is the primary output of the constraint solver.

8.3 Data Flow Pipeline

The planning pipeline executes in three phases:

1. **Collect.** Pull tasks from OmniFocus (via OmniFocusReader), calendar events from EventKit, and health metrics from HealthKit and the HRV sensor.
2. **Model.** Compute the day's energy budget and slot types using the circadian model modulated by physiological state. Apply weekly constraints (taper, recovery, pacing).
3. **Assign.** Run the constraint solver to place tasks into compatible slots. Optionally pass the draft plan through the LLM planner for natural-language refinement and common-sense checks.

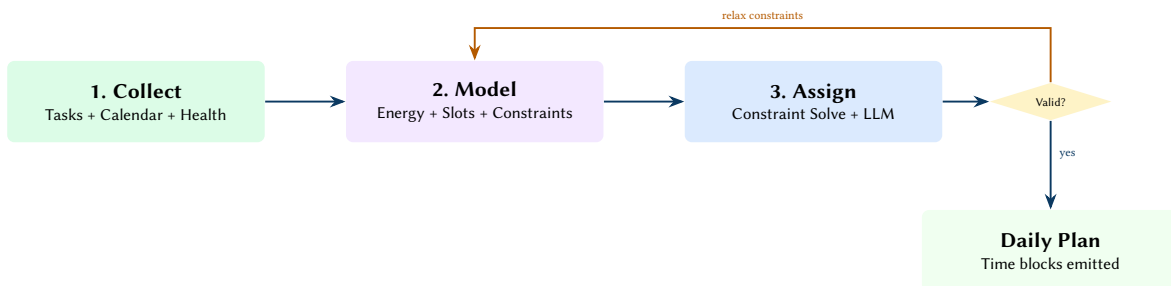


Figure 17. Three-phase planning pipeline with constraint relaxation feedback loop. If no valid assignment exists, the solver relaxes lower-priority constraints and retries.

9 Constraint Solver

The solver assigns tasks to slots in a priority-ordered greedy pass with backtracking.

9.1 Assignment Algorithm

The solver processes tasks in priority order:

1. **Fixed events first.** Calendar appointments are placed as immovable blocks, marking their slots as unavailable.
2. **Deadline-driven tasks.** Tasks with today's deadline are placed next, in earliest-compatible-slot order.
3. **Deep work.** Remaining Deep tasks fill Peak slots, respecting the willpower drain model and 90-minute batching preference.
4. **Physical activities.** Placed in Sustain or Peak slots, with mandatory mobility windows.
5. **Shallow tasks.** Fill remaining Sustain and Trough slots.

Insight:
A greedy solver with constraint relaxation is simpler and more transparent than a full CSP solver, and produces “good enough” plans that the user can manually adjust.

6. **Recovery.** Any unfilled slots receive Recovery type.

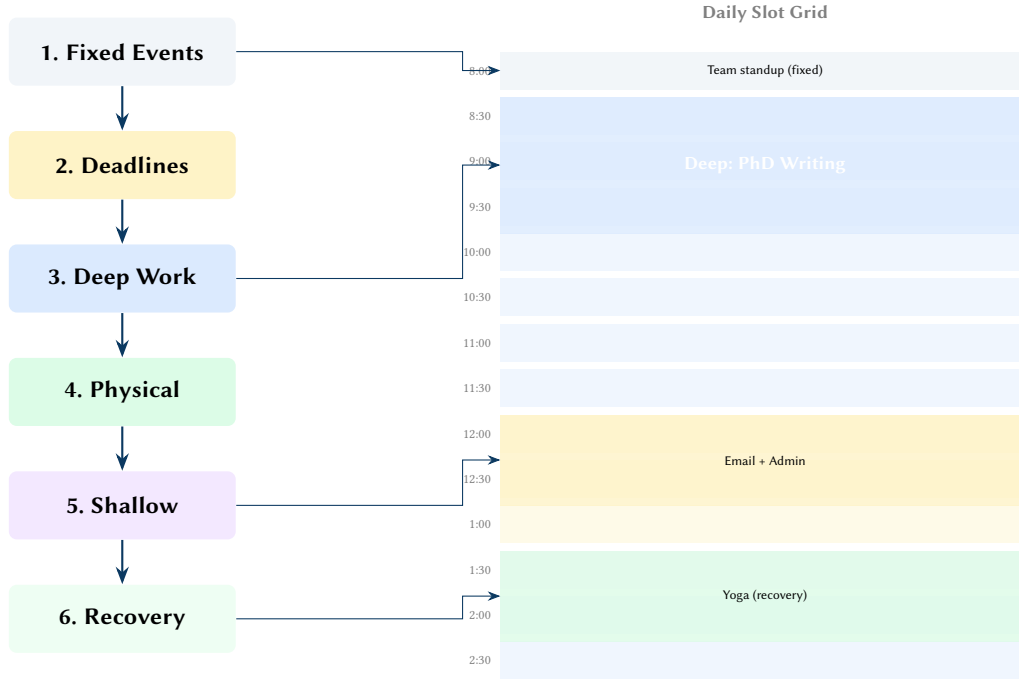


Figure 18. Greedy assignment algorithm. Tasks are placed in priority order (left) into the daily slot grid (right). Each task type targets compatible capacity zones.

9.2 Constraint Relaxation

When the solver cannot place all deadline-driven tasks, it relaxes constraints in a defined order:

1. Allow Deep tasks in Sustain slots (mild relaxation).
2. Reduce minimum deep-work block size from 90 to 60 minutes.
3. Allow shallow/deep interleaving (context-switch penalty accepted).
4. Extend the day by 30-minute increments (up to user-defined limit).
5. Flag the task as *unschedulable* and alert the user.

Each relaxation is logged so the user can see why the plan deviates from ideal.

10 Task Clustering and Context-Switch Minimisation

Beyond placing individual tasks in compatible slots, **WP** actively clusters related tasks to minimise context-switching costs. Research by Mark et al. found that after an interruption, it takes an average of 23 minutes to return to full focus on the original task. Monsell’s work on task switching established that “switch costs” are robust even when the switch is predictable and preparation time is available.

Insight:
Transparent constraint relaxation builds trust—the user sees why a deep-work block landed in the afternoon, not just that it did.

10.1 Clustering Dimensions

Tasks can be clustered along several orthogonal dimensions. The solver considers multi-dimensional affinity when assigning tasks to adjacent slots:

10.2 Context-Switch Penalty

The solver incorporates a context-switch penalty term into the assignment objective. For adjacent slots h_i and h_{i+1} assigned to tasks τ_a and τ_b :

$$\text{Penalty}(h_i, h_{i+1}) = w_{\text{switch}} \cdot \Delta(\tau_a, \tau_b)$$

where Δ measures the “distance” between task contexts across all clustering dimensions. Switching from deep coding to deep writing incurs a low penalty; switching from deep coding to an errand incurs a high one.

Insight:
Context-switching is not just a nuisance—it is a measurable cognitive tax. The solver should minimise the number of switches, not just the type of work in each slot.

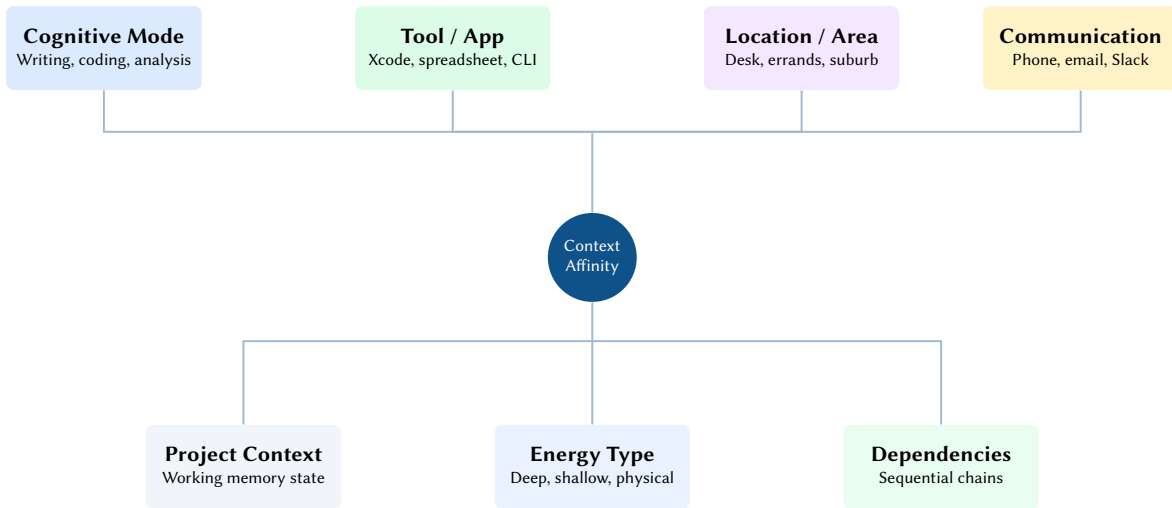


Figure 19. Seven clustering dimensions converge into a context-affinity score. Tasks sharing multiple dimensions receive high affinity and are scheduled in adjacent slots to minimise context-switching overhead.

: Batching Preference

The solver prefers to *extend* an existing block of the same task type rather than interleave different types. A 3-hour deep-work block is preferred over three 1-hour blocks separated by shallow tasks, even if capacity permits interleaving.

10.3 Errand Clustering

Location-dependent tasks (errands, appointments at specific venues) are clustered into a single outing block with an optimised route. This maps to a variant of the Travelling Salesman Problem, which for typical errand counts (< 15) admits exact solutions trivially.

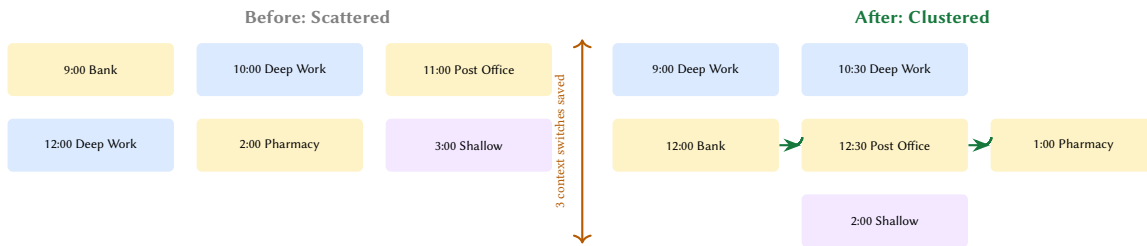


Figure 20. Errand clustering transforms scattered location-based tasks into a single outing block with an optimised route, saving three context switches and eliminating redundant travel time.

11 Smart Prioritisation

WP uses a multi-factor priority score that goes beyond simple deadline urgency. Drawing from operations research (Weighted Shortest Job First) and multi-criteria decision analysis, the composite priority is:

$$P(\tau_j) = \sum_k w_k \cdot f_k(\tau_j)$$

where each factor f_k captures a distinct dimension:

11.1 Urgency: Sigmoid Escalation

Rather than a binary due/not-due flag, urgency follows a sigmoid that begins climbing ~5 days before the deadline and saturates at the deadline itself:

$$f_{\text{urgency}}(\tau_j) = \frac{1}{1 + e^{-k(t - \delta(\tau_j) + b)}}$$

This produces “gentle escalation” rather than last-minute panic, giving the solver time to fit the task into a compatible slot before crisis mode.

Insight:
Sigmoid urgency prevents the common failure of ignoring a task until the deadline day, then cramming it into a Trough slot because no Peak slots remain.

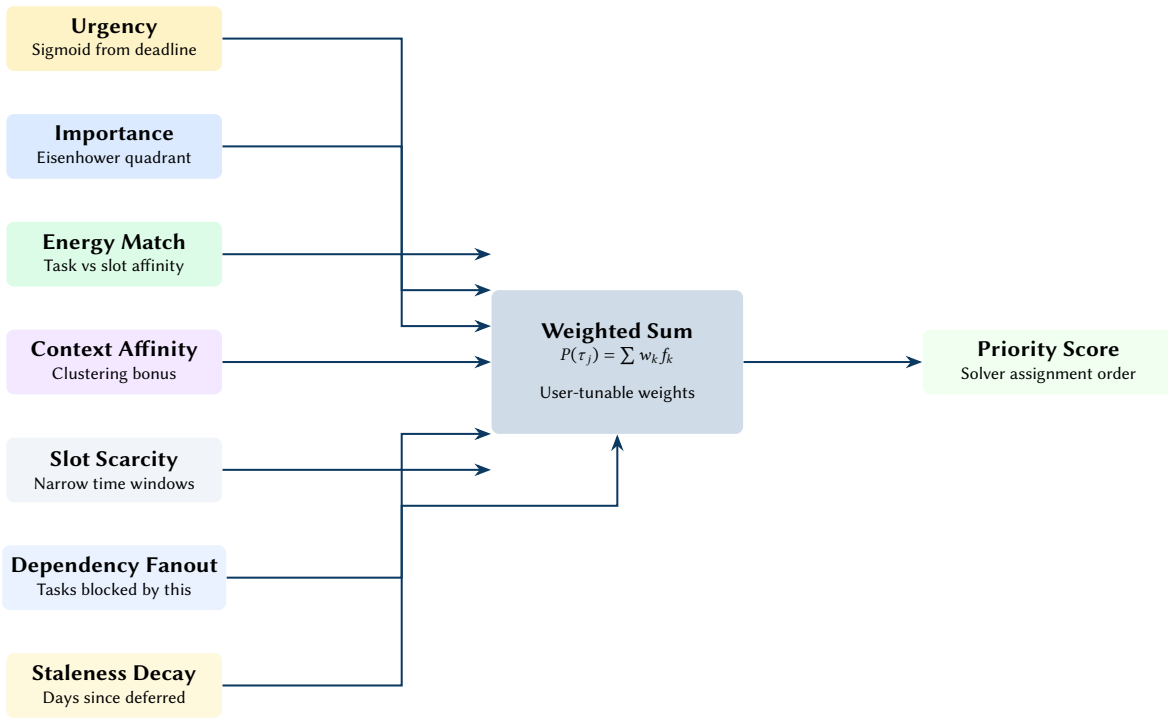


Figure 21. Multi-factor priority scoring. Seven weighted factors converge into a composite score that determines solver assignment order. Weights are user-tunable during onboarding.

11.2 Slot Scarcity

A task that can only be done during a narrow window (bank errand during business hours, phone call in a colleague’s timezone) receives a priority boost proportional to the scarcity of compatible slots:

$$f_{\text{scarcity}}(\tau_j) = \frac{1}{|\text{compatible_slots}(\tau_j)| + 1}$$

If there are only 2 compatible 30-minute slots today for a bank errand, it is more urgent to schedule than a task with 12 compatible slots.

11.3 The Commitment Budget

At the weekly level, when committed tasks exceed available capacity, the system quantifies the overcommitment:

: Overcommitment Alert

If $\sum_j d(\tau_j) > \sum_d \text{usable_slots}(d)$ for the week, WP alerts the user with: “You have X hours of work committed to a week with Y hours of capacity. Here are the N lowest-priority tasks you could defer.”

Urgency Sigmoid vs Binary Threshold

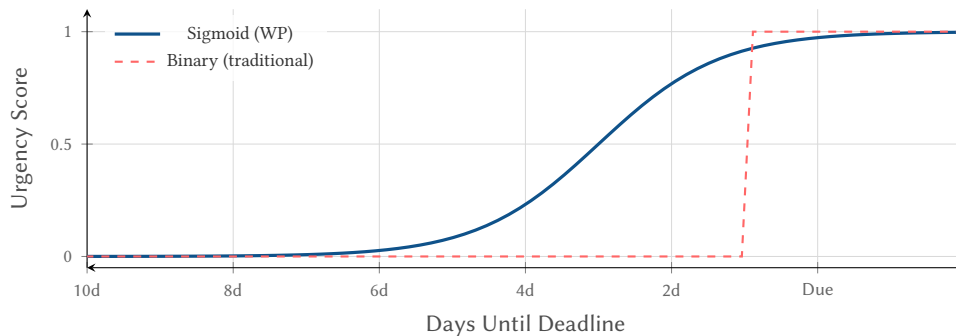


Figure 22. Sigmoid urgency vs binary threshold. The sigmoid begins escalating 5 days before deadline, providing gradual pressure. The binary model ignores the task until the due date, then jumps to maximum urgency.

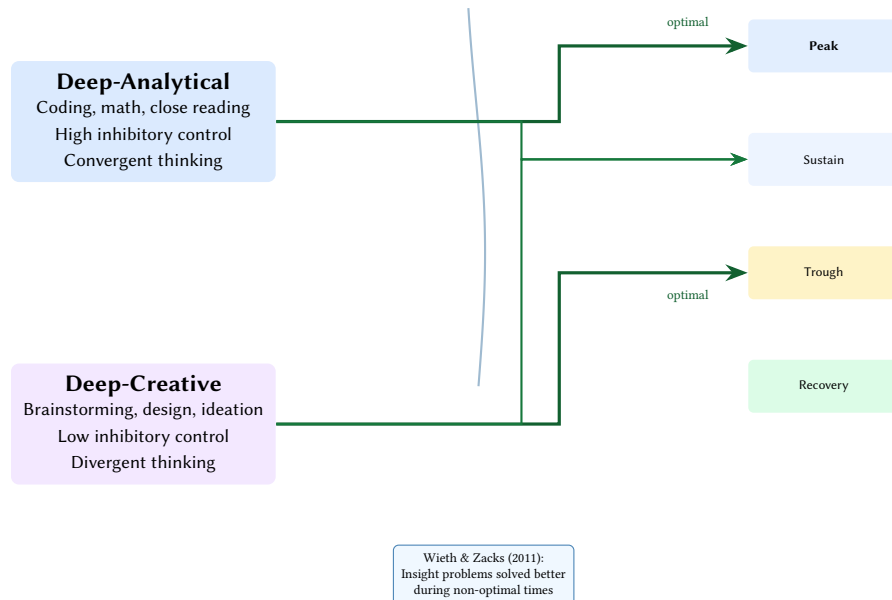
12 Extended Feature Set

Beyond the core scheduling model, **WP** targets a set of extended features that differentiate it from existing tools. A competitive analysis of Sunsama, Motion, Reclaim.ai, Akiflow, Structured, and Sorted3 reveals a clear gap: no existing product combines automatic scheduling with physiological awareness. This section details each extended feature in depth, including the science, data model extensions, and constraint implications.

12.1 Deep-Creative vs Deep-Analytical

Research by Wieth and Zacks (2011) found a counter-intuitive result: *insight* problems (creative, lateral thinking) are solved better during *non-peak* circadian times. Reduced inhibitory control during non-peak periods allows broader associative thinking.

WP therefore sub-classifies Deep work into two modes with distinct scheduling affinities:



Insight:
This challenges the simple Deep = Peak mapping. Creative brainstorming may actually benefit from Sustain or even Trough slots.

Figure 23. Deep work sub-classification. Analytical work maps to Peak slots (high inhibitory control needed), while Creative work maps optimally to Sustain/Trough slots where reduced inhibition enables divergent thinking.

The task classifier (Section 7) is extended with tag patterns: tasks tagged “brainstorm”, “ideate”, “design exploration” or in creative projects map to Deep-Creative; tasks tagged “coding”, “analysis”, “review” map to Deep-Analytical.

12.2 Ultradian Rhythm Timer

Nathaniel Kleitman’s Basic Rest-Activity Cycle (BRAC) suggests the brain cycles between high and low alertness every ~90 minutes. **WP** aligns focus blocks with this ultradian rhythm rather than arbitrary Pomodoro intervals:

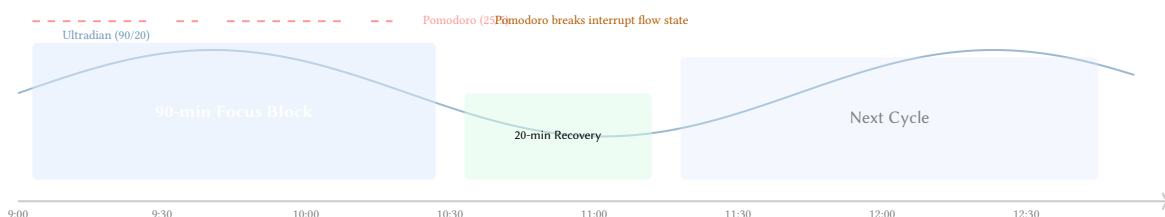


Figure 24. Ultradian rhythm alignment. A 90-minute focus block followed by a 20-minute recovery matches Kleitman’s BRAC, unlike Pomodoro’s arbitrary 25/5 cadence which interrupts flow states mid-cycle.

Within a 90-minute block, the timer offers optional micro-cues:

- **30-minute mark:** brief posture check (stand, stretch neck).
- **60-minute mark:** 20-20-20 eye rule (look 20 feet away for 20 seconds).
- **90-minute mark:** full break—walk, hydrate, change environment.

12.3 Habit Tracking with Physiological Feedback

Habits are modelled as recurring tasks with two special properties: (1) they have preferred time windows, and (2) their completion feeds back into the energy model.

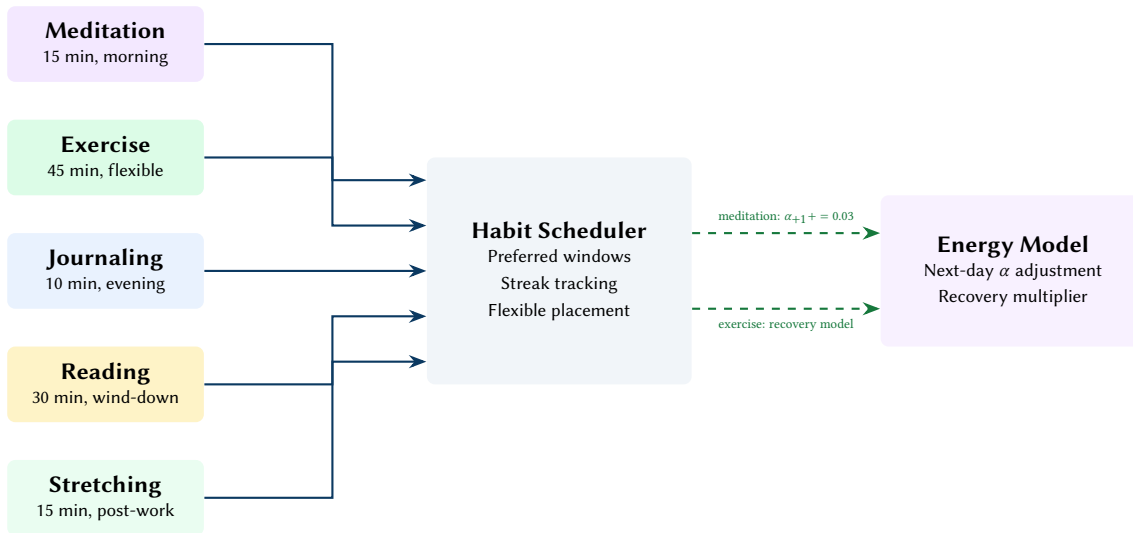


Figure 25. Habit tracking with energy feedback. Habits are scheduled in preferred time windows; their completion modifies next-day energy parameters. Morning meditation improves α by +0.03; exercise triggers the full recovery model.

Table 5. Habit physiological feedback effects.

Habit	Preferred Window	Feedback	Effect
Morning meditation	6:00–7:30	$\alpha_{+1} + = 0.03$	Improved next-day HRV
Journaling	20:00–21:00	$\beta_{+1} + = 0.02$	Better sleep onset
Evening reading	21:00–22:00	$\beta_{+1} + = 0.01$	Reduced screen time
Yoga / stretching	17:00–18:00	$\rho + = 0.05$	Active recovery boost
Exercise	Flexible	Recovery model	Taper/recovery cascade

12.4 Social and Family Calendar Awareness

Family commitments create hard constraints that override all other scheduling logic. **WP** reads shared iCloud calendars and classifies events by constraint type:

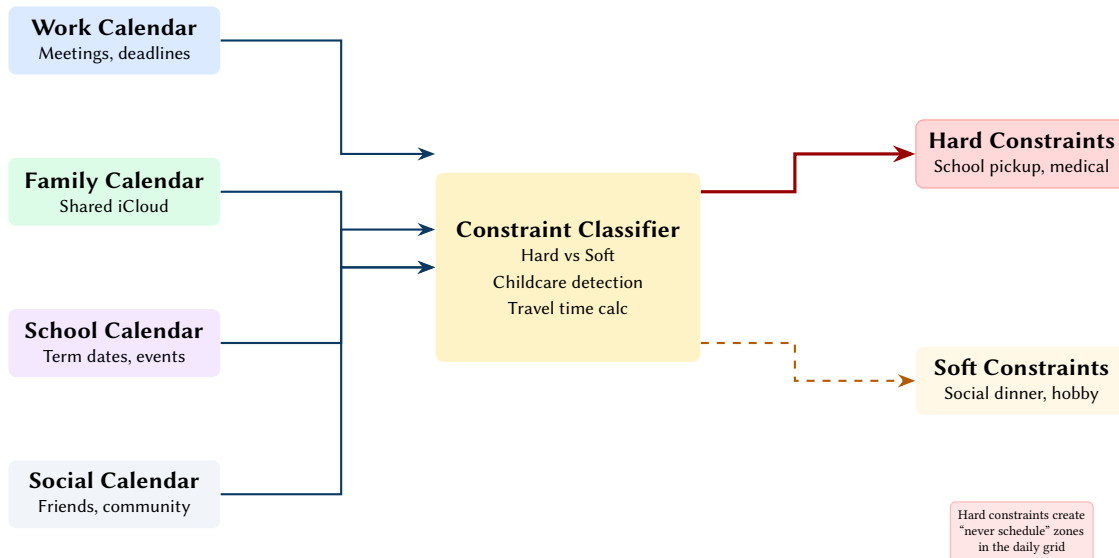


Figure 26. Social and family calendar integration. Events from shared calendars are classified as hard constraints (non-negotiable, block slots entirely) or soft constraints (prefer to avoid, but can override if urgent work demands it).

: Family-First Invariant

No task, regardless of priority or deadline, may be scheduled in a slot occupied by a hard family constraint. The system may suggest renegotiating the family commitment via user notification, but will never silently override it.

12.5 Meal Timing and Cognitive Performance

The post-lunch dip is already modelled in the circadian curve, but meal *timing* and *composition* can amplify or attenuate it. Wells et al. (1998) showed that high-carbohydrate lunches deepen the dip while protein-rich meals attenuate it.

WP adds meal-awareness constraints:

- **Lunch placement:** scheduled *after* the morning deep-work block ends, never during it.
- **Pre-exercise buffer:** main meals must precede high-intensity physical activity by ≥ 120 minutes.
- **Caffeine cutoff:** no caffeine after 14:00 (configurable) to protect sleep quality and tomorrow's β factor.
- **Post-lunch slot typing:** the 60 minutes following lunch are automatically downgraded from Sustain to Trough if a heavy meal is indicated.

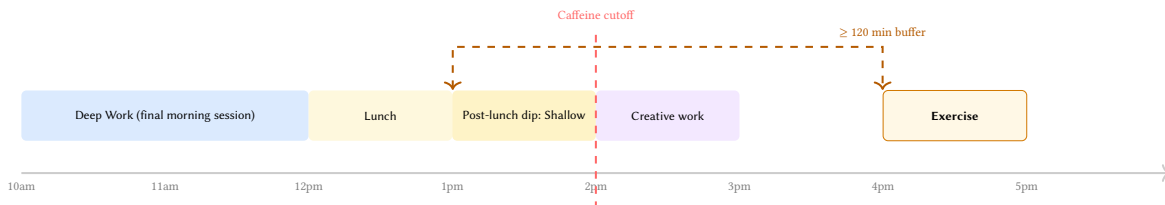


Figure 27. Meal timing constraints. Lunch is placed after the deep-work block; a 120-minute buffer separates meals from exercise; caffeine has a hard cutoff to protect sleep. The post-lunch slot is automatically downgraded.

12.6 Weather-Aware Scheduling

Outdoor tasks (errands, walks, outdoor exercise, garden work) should be scheduled on days and times with favourable weather. Integration with Apple WeatherKit provides hourly and 7-day forecasts.

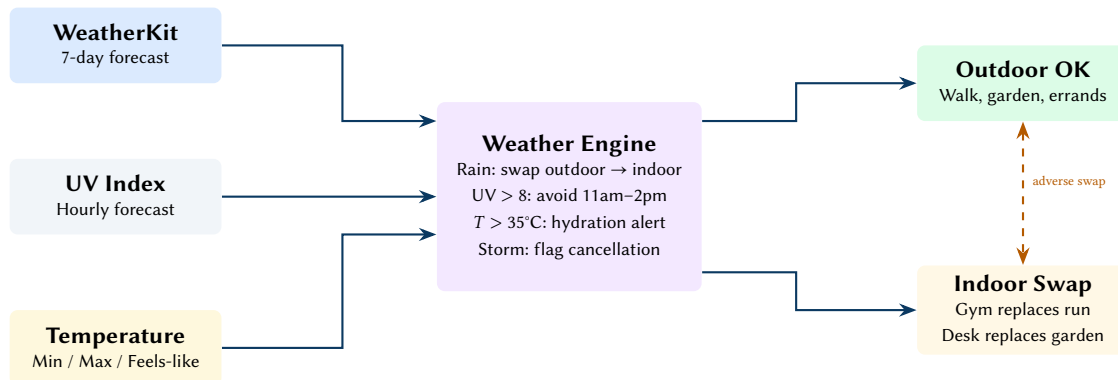


Figure 28. Weather-aware scheduling with multi-signal input. Temperature, UV index, and precipitation forecasts drive outdoor/indoor task swaps, UV-avoidance windows, and extreme-heat alerts specific to Adelaide's climate.

Adelaide-specific considerations: UV index regularly exceeds 11 in summer, making outdoor activity between 11 am and 2 pm inadvisable. The system automatically shifts outdoor walks and exercise to early morning or late afternoon during high-UV seasons.

12.7 Travel Time Estimation

When consecutive tasks require different locations, WP inserts travel blocks using Apple MapKit with time-of-day traffic awareness:

Table 6. Travel mode classification and energy impact.

Mode	Duration Factor	Energy Type	Productivity
Walking (< 20 min)	1.0×	Recovery (positive)	None
Cycling	1.0×	Moderate physical	None
Driving	1.0×	Low cognitive drain	None
Public transit	1.0–1.5×	Minimal	Shallow tasks OK

12.8 Meeting Preparation and Debrief

Meetings are not standalone events—they require cognitive ramp-up before and action capture after. WP automatically inserts buffer blocks:

Insight:
Transit time is not dead time. Public transport blocks are marked as “shallow-compatible”, creating found time for email, reading, or podcast listening—effectively converting travel into productive shallow work.

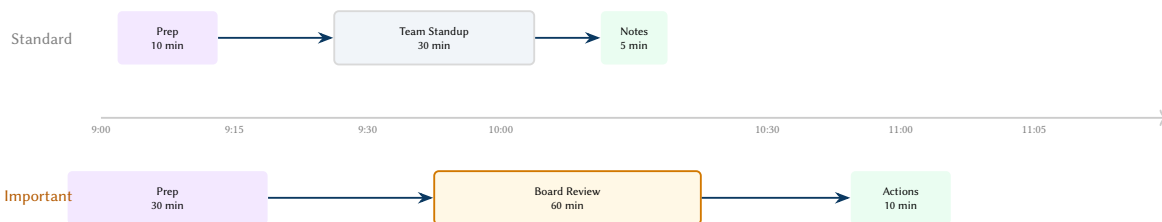


Figure 29. Automatic meeting buffers. Standard meetings receive 10-minute prep and 5-minute debrief; important meetings (board, review, presentation) receive 30-minute prep and 10-minute action capture.

12.9 Focus Mode and Notification Integration

WP maps time block types to macOS Focus profiles, creating an automated attention management layer:

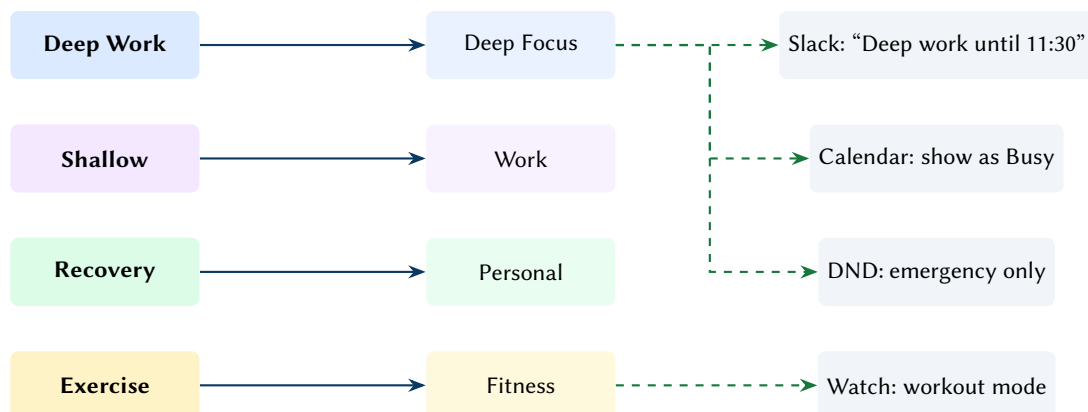


Figure 30. Focus mode integration cascade. Each block type triggers a Focus profile which in turn configures Slack status, calendar visibility, DND settings, and Apple Watch mode. Deep Work triggers the full notification suppression chain.

12.10 Energy Forecasting from Wearable Trends

Rather than using only this morning's HRV to modulate today, WP uses multi-day trends to forecast the week's capacity during the Sunday review. The forecasting model uses three signal inputs:

1. **HRV trend line:** 30-day EWMA of morning RMSSD, with day-of-week seasonality (HRV typically drops Mondays, climbs mid-week).
2. **Training load:** 7-day rolling average of physical energy expenditure. High training load predicts lower next-day HRV.
3. **Sleep staging:** Apple Watch sleep stages (watchOS 9+) provide REM and deep-sleep percentages. Nights with < 15% deep sleep predict reduced next-day cognitive capacity.

7-Day HRV Trend with Energy Forecast

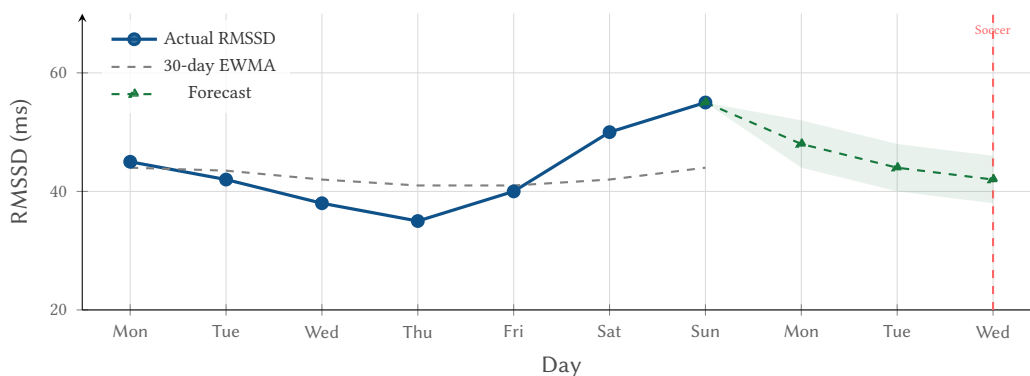


Figure 31. Energy forecasting from HRV trends. The past week's RMSSD values (solid) and the 30-day EWMA baseline (dashed gray) inform a 3-day forecast (dashed green with confidence band) used during the weekly review to pre-allocate deep work to high-capacity days.

Insight:
Proactive energy forecasting enables the weekly review to pre-assign deep work to the days most likely to support it, rather than discovering a depleted state each morning.

12.11 Duration Calibration and Planning Fallacy Correction

Kahneman’s research on the planning fallacy [11] shows most people underestimate task durations by $\approx 30\text{--}50\%$. **WP** combats this by tracking planned-vs-actual durations for each task type and building per-user correction factors.

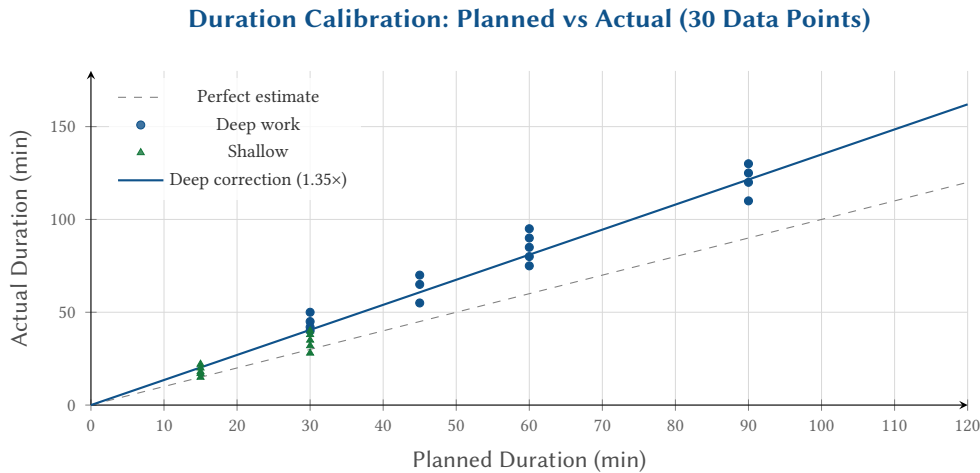


Figure 32. Duration calibration. Scatter plot of planned vs actual durations reveals a consistent 35% underestimation for deep work. After 30 data points, the system applies a 1.35 $\times$ correction factor automatically.

The calibration model operates per-task-type:

- **Cold start** (< 10 data points): use the user’s estimate with a warning (“This is an uncalibrated estimate”).
- **Warming** ($10\text{--}30$ points): blend user estimate with historical average, weighted by confidence.
- **Calibrated** (≥ 30 points): use the per-type correction factor $\mu_{\text{actual}}/\mu_{\text{planned}}$.

12.12 Training Periodisation

For users with structured training plans (running programmes, strength cycles, sport seasons), **WP** can import the training calendar and automatically derive taper/recovery constraints from prescribed intensity:

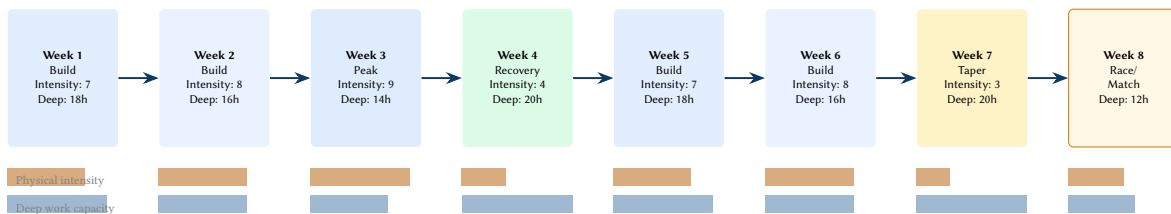


Figure 33. Training periodisation over an 8-week mesocycle. Physical intensity (orange bars) is inversely correlated with deep-work capacity (blue bars). Recovery weeks and taper weeks automatically expand the cognitive work budget.

: Periodisation Coupling

When a training plan specifies a peak-intensity week, the weekly deep-work cap is automatically reduced by $\text{intensity} \times 2$ hours. During recovery weeks, the cap is restored to its full value.

12.13 Seasonal and Light Exposure Adjustment

Adelaide’s latitude (34.9°S) produces significant seasonal variation: ~ 10 hours of daylight in winter vs ~ 14.5 hours in summer. This affects circadian phase through the **SCN**’s light-dependent entrainment pathway.

WP adjusts the circadian phase offset ϕ_0 monthly based on sunrise times, and can recommend:

- **Winter:** morning light exposure (outdoor walk within 30 min of waking) to advance the circadian clock. Light therapy lamp as alternative on overcast days.
- **Summer:** evening blue-light reduction (Night Shift, f.lux) to prevent delayed sleep onset.
- **Transition months:** gradual shift of deep-work blocks to track the moving Peak window.

12.14 Plan Quality Learning

Over time, **WP** tracks a “plan quality score” for each day: the percentage of scheduled tasks completed in their assigned slots. This score is correlated with observable variables to build a predictive model:

***Insight:**
After 60+ days of data, the model can proactively warn: “Days with > 3 meetings have historically scored 40% lower on plan adherence. Today has 4 meetings—suggesting a lighter task load.”*

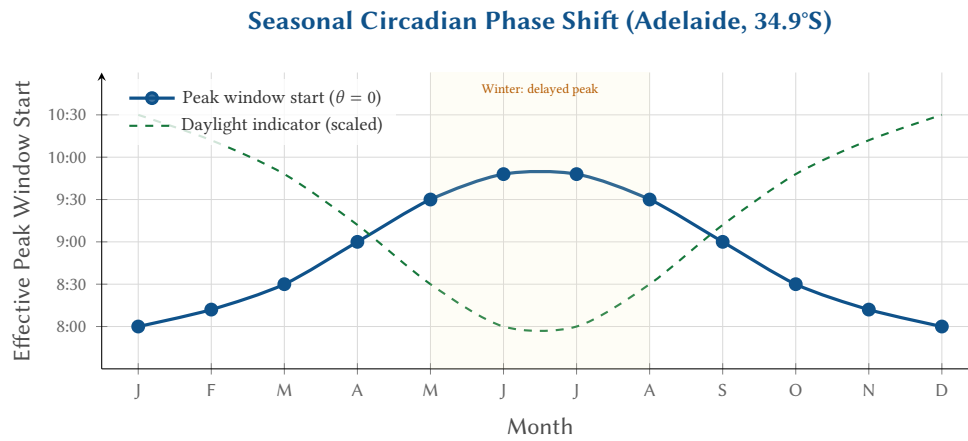


Figure 34. Seasonal phase shift of the cognitive Peak window. In Adelaide’s winter (May–Aug), reduced morning light delays the effective Peak start by ~1.5 hours. The system adjusts ϕ_0 seasonally using sunrise data.

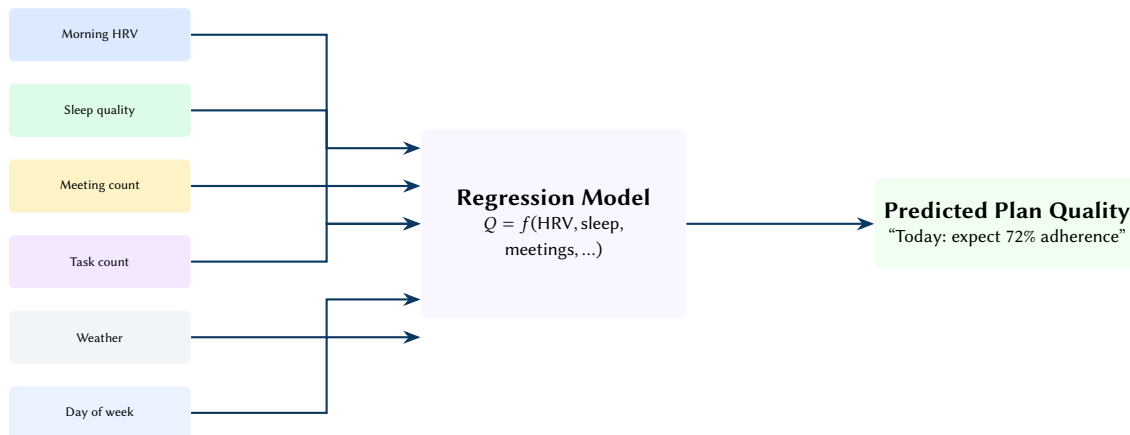


Figure 35. Plan quality prediction model. Six observable factors feed a regression that predicts today’s likely adherence score. High-meeting days historically score ~40% lower; the system responds by planning lighter task loads on meeting-heavy days.

12.15 Competitive Gap

Insight:
The competitive gap is clear: no existing product combines automatic scheduling with physiological awareness. This is WP’s primary differentiator—a planner that knows not just what you need to do, but what you are capable of doing right now.

Table 7. Competitive feature matrix. ✓ = present, – = absent.

Feature	Sunsama	Motion	Reclaim	Sorted3	WP
Auto-scheduling	–	✓	Partial	Basic	✓
Circadian awareness	–	–	–	–	✓
HRV / health integration	–	–	–	–	✓
Recovery modelling	–	–	–	–	✓
Energy budgeting	Crude	–	–	–	✓
Context clustering	–	–	–	–	✓
Deep-Creative splitting	–	–	–	–	✓
Habit feedback loop	–	–	Basic	–	✓
Mid-day reflow	–	✓	Partial	–	✓
Weekly pacing	–	–	–	–	✓
Training periodisation	–	–	–	–	✓
Weather awareness	–	–	–	–	✓
Seasonal adjustment	–	–	–	–	✓
Duration calibration	–	–	–	–	✓
Family constraints	–	–	–	–	✓
Focus mode cascade	–	–	Partial	–	✓
Transparent reasoning	–	–	–	–	✓
Apple native	App	Web	Web	✓	✓

13 Interaction Model

WP supports three interaction loops aligned with Cal Newport’s planning methodology [9]:

13.1 Weekly Review (Sunday Evening)

The weekly review generates a 7-day skeleton:

1. Pull all tasks with due dates in the coming week from OmniFocus.
2. Overlay fixed calendar commitments (meetings, appointments).
3. Identify high-intensity events and propagate taper/recovery constraints.
4. Allocate deep-work blocks to Peak slots on non-recovery days.
5. Present the weekly skeleton for user review and adjustment.

13.2 Morning Plan (Post-HRV Reading)

The morning plan takes the weekly skeleton, adjusts for actual recovery state (morning HRV reading), and produces the final daily time-block schedule.

13.3 Daily Shutdown (Evening)

At day’s end, the system captures what was actually completed vs planned, feeding the history store for future calibration.

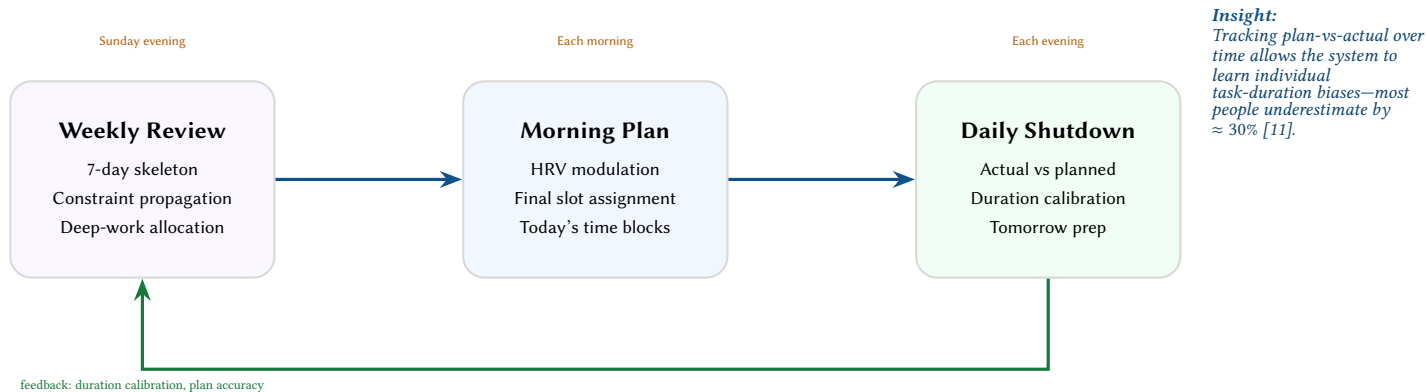


Figure 36. Three interaction loops forming the planning cycle. The weekly review sets the skeleton, morning planning finalises each day, and the shutdown ritual captures actuals for continuous calibration.

13.4 Mid-Day Reflow

When reality diverges from the plan—a meeting overruns, an urgent task arrives, or energy crashes earlier than predicted—the system offers a reflow:

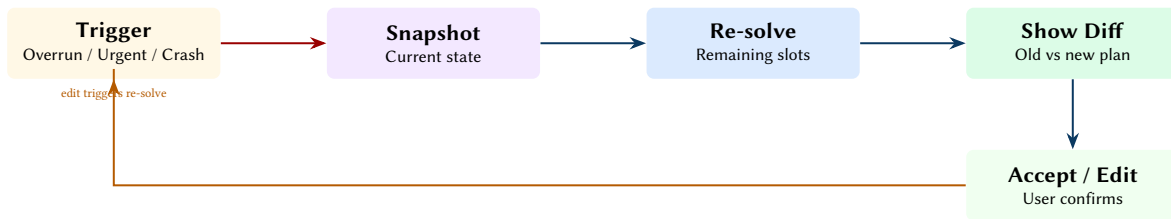


Figure 37. Mid-day reflow pipeline. A disruption triggers a snapshot of current progress, re-solves remaining slots, and presents a diff for user approval.

14 macOS Integration

WP targets macOS as the primary platform, leveraging native frameworks for seamless integration with the Apple ecosystem.

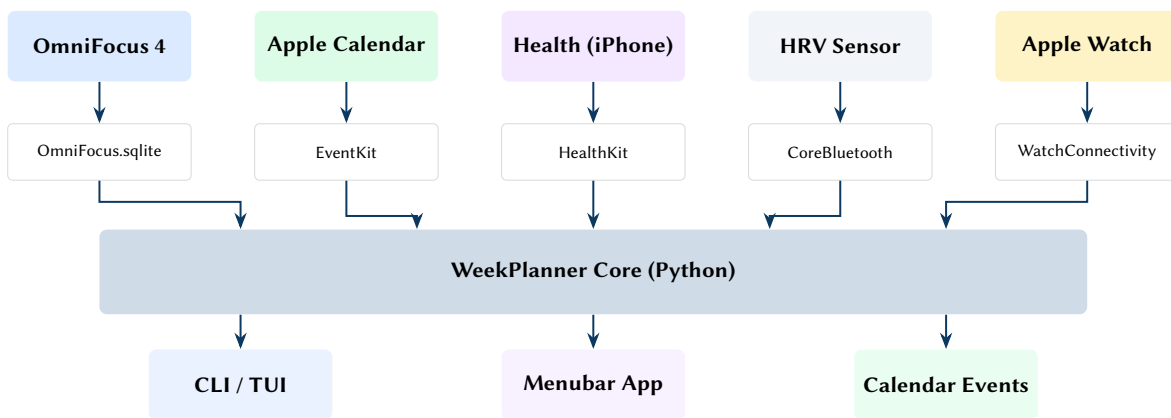


Figure 38. macOS integration map. Five data sources feed through Apple frameworks into the WeekPlanner core, which emits plans to three output channels: CLI/TUI for power users, menubar for glanceable status, and Calendar write-back for visual time blocking.

15 Implementation Plan

Development is phased to deliver value incrementally:

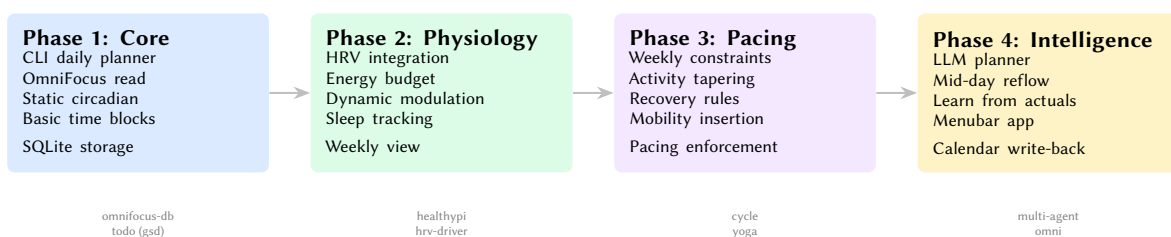


Figure 39. Phased implementation roadmap with ecosystem dependencies. Each phase builds on the previous and draws from specific existing projects.

Design Decision 1: Technology Choice: Python First

Phase 1 targets Python to maximise reuse of omnifocus-db, healthypi, and omni components. Performance-critical paths (HRV signal processing) remain in Rust via the existing hrv-driver and cardio-rs crates, exposed to Python through Parquet interchange or optional PyO3 bindings.

16 Example: A Concrete Day

To ground the abstractions, consider a concrete Wednesday—the day of a 7pm soccer match—planned for an intermediate chronotype with a slightly below-baseline morning HRV ($\alpha = 0.85$, $\beta = 1.0$).

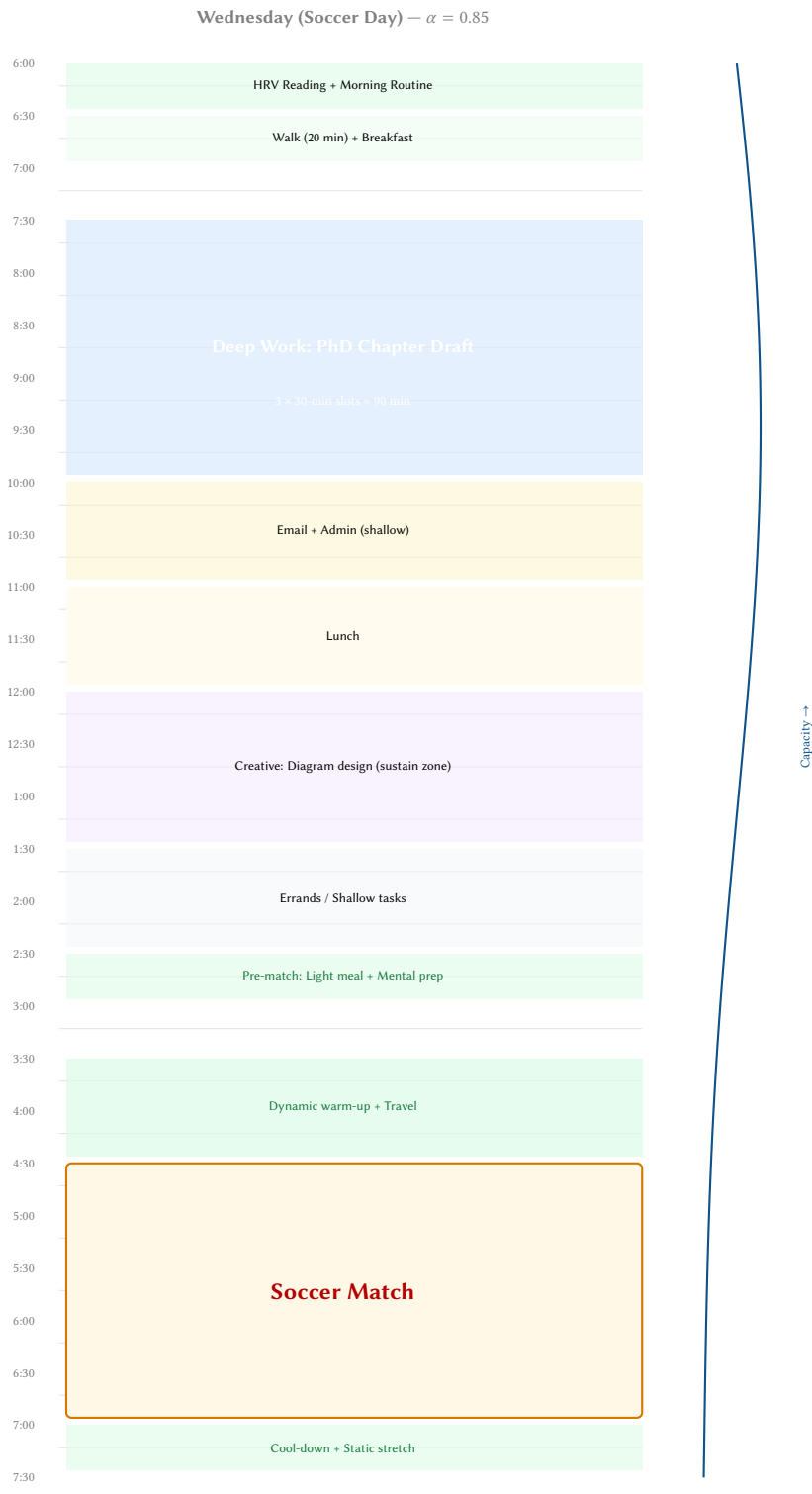


Figure 40. A complete daily plan for a soccer day. Deep work is front-loaded in the Peak window; the afternoon transitions through shallow tasks to pre-match preparation. Mandatory warm-up and cool-down blocks bracket the match. The right-side curve shows the modulated circadian capacity.

17 Sleep Architecture and Next-Day Performance

Sleep is not a monolithic state—its internal architecture directly determines next-day cognitive capacity. WP ingests Apple Watch sleep staging data (available since watchOS 9) to compute a fine-grained sleep quality score that modulates β more precisely than total sleep time alone.

17.1 Sleep Stage Impact Model

17.2 Sleep Debt Accumulation

Sleep debt compounds non-linearly. Lim and Dinges’ meta-analysis showed that even partial deprivation (4–6 hours) significantly impairs sustained attention with large effect sizes ($d = 0.74$ after one night).

Insight:
Truncating sleep by waking early primarily sacrifices REM, which disproportionately impairs creative and emotional capacity. On short-sleep nights, WP should preferentially schedule analytical work over creative tasks.

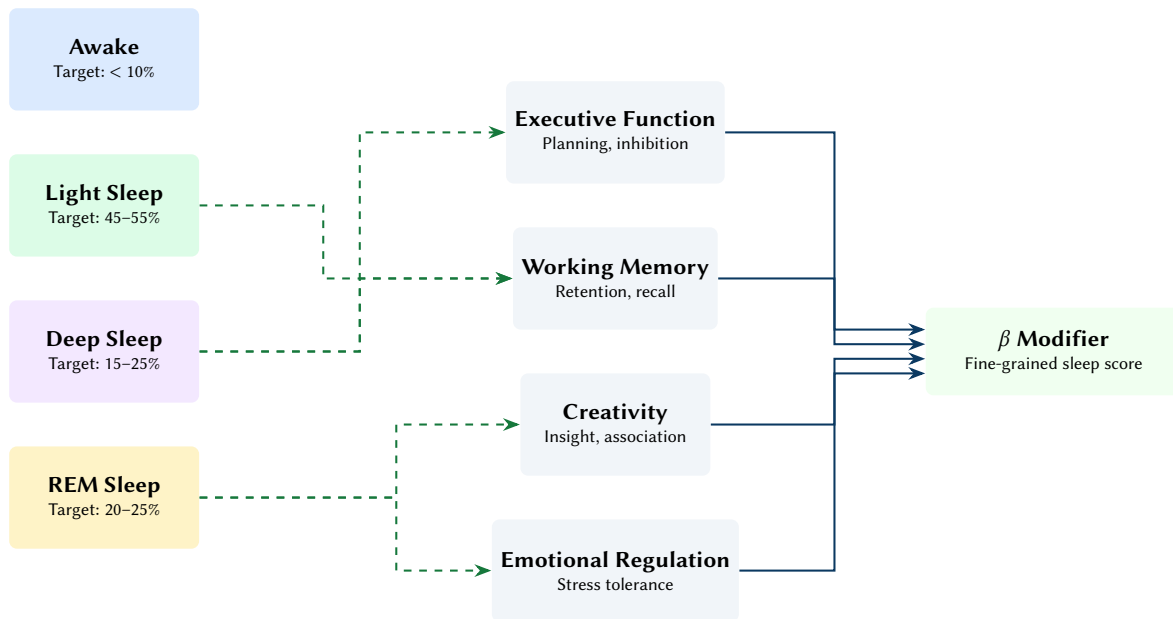


Figure 41. Sleep stage impact model. Deep sleep primarily supports executive function and memory consolidation; REM sleep supports creativity and emotional regulation. Each stage’s proportion relative to target modulates the next-day β factor.

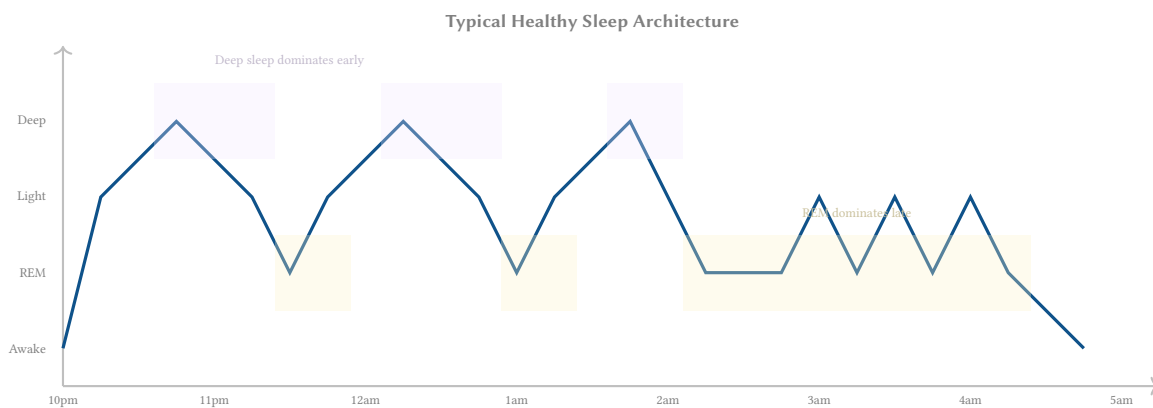


Figure 42. Hypnogram showing typical sleep architecture. Deep sleep dominates the first half of the night (critical for executive function); REM sleep dominates the second half (critical for creativity and emotional regulation). Truncating sleep primarily sacrifices REM.

: Sleep Emergency Protocol

If $k \geq 5$ consecutive nights below 7 hours, **WP** enters “sleep emergency” mode: all deep work is suspended, the day is planned as Recovery-only, and a mandatory early-bedtime block is inserted.

Cumulative Sleep Debt and Cognitive Capacity

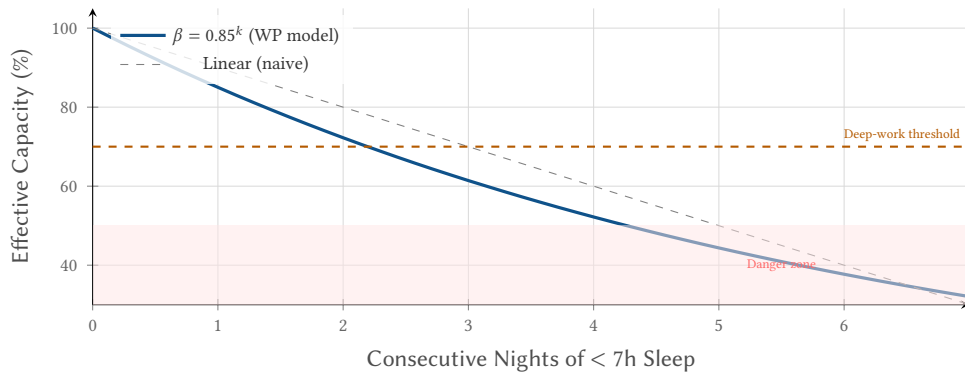


Figure 43. Cumulative sleep debt effect on cognitive capacity. The exponential model (0.85^k) drops below the deep-work threshold after just 2 consecutive short nights. After 5 nights, capacity is below 45%—the system switches to Recovery-only planning.

18 Caffeine Pharmacokinetics

Caffeine’s half-life of ~5–6 hours means a 2 pm coffee still has 50% active concentration at 7–8 pm, directly impacting sleep onset and quality. **WP** models caffeine as a scheduling constraint.

Caffeine Concentration Over Time

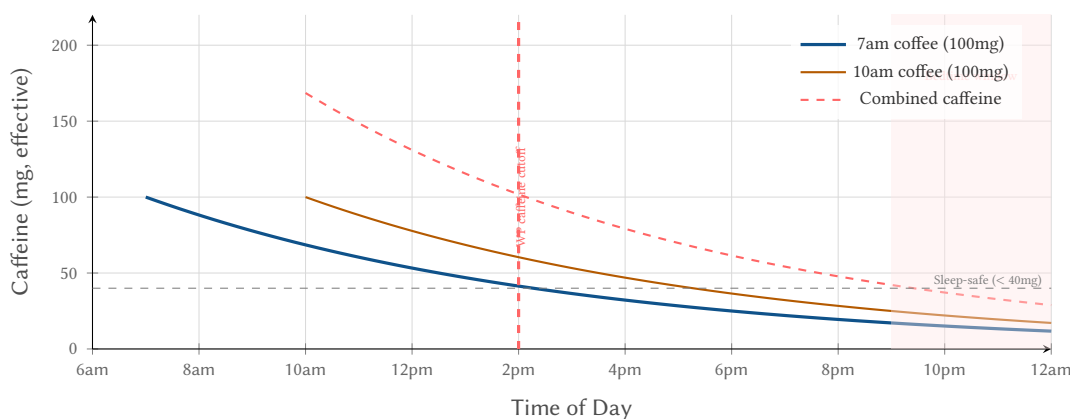


Figure 44. Caffeine pharmacokinetics. A 7 am coffee decays to safe levels by bedtime, but a 2 pm coffee still has ~50 mg active at 10 pm. The WP cutoff at 2 pm (configurable) ensures caffeine clears before the target bedtime.



Figure 45. Caffeine timing zones. The optimal window (90 minutes post-wake) aligns with the natural cortisol dip. The hard block after 2 pm protects tonight’s sleep quality and tomorrow’s β factor.

19 The Shutdown Ritual

Cal Newport’s “shutdown complete” protocol is a cornerstone of sustainable productivity [2]. **WP** formalises this as a guided evening workflow that produces both closure for today and preparation for tomorrow.

19.1 Cognitive Closure

The ritual’s primary purpose is psychological, not logistical. By reviewing every open loop and confirming that each has a capture point (either completed, deferred with a date, or added to OmniFocus), the Zeigarnik effect—the tendency of incomplete tasks to occupy working memory—is neutralised.

Insight:
The common habit of “afternoon coffee to push through the slump” is counter-productive in a multi-day model: it borrows energy from tomorrow’s capacity. **WP** suggests a 10-minute walk instead, which restores alertness without caffeine’s sleep penalty.

Insight:
Without shutdown closure, open work loops intrude on evening recovery time, degrading sleep quality and thus tomorrow’s β . The ritual is not optional overhead—it is a direct input to the energy model.

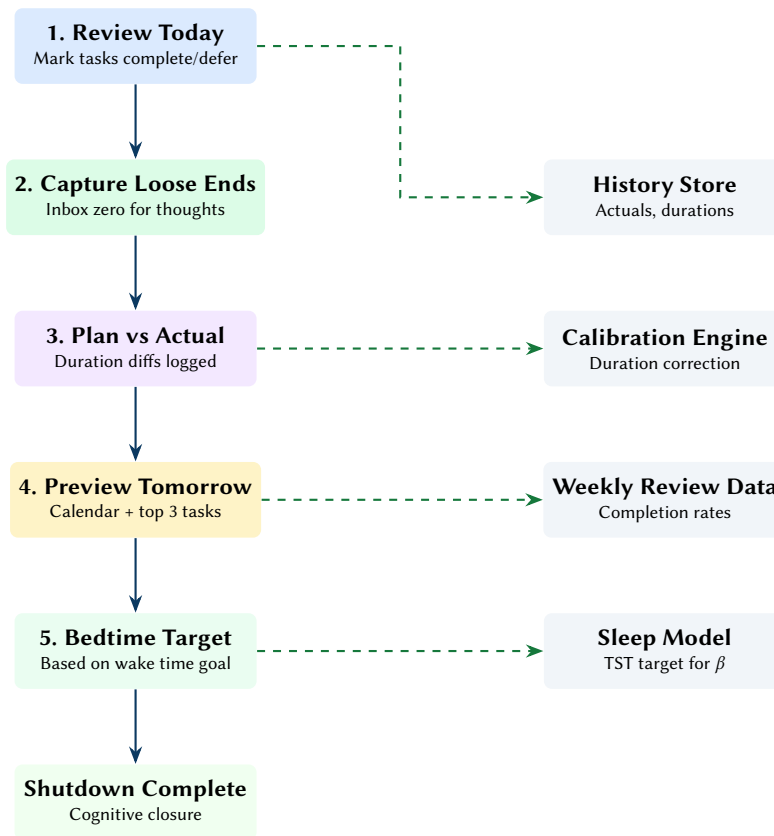


Figure 46. The shutdown ritual as a 5-step guided workflow. Each step produces data that feeds back into the planning system: duration calibration, weekly completion metrics, and sleep targets.

20 Onboarding and Chronotype Calibration

WP requires initial calibration of several personal parameters. The onboarding flow gathers these through a combination of questionnaire and observation.

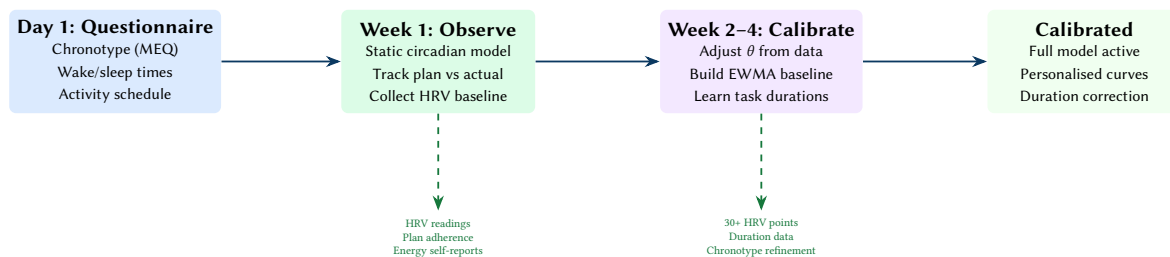


Figure 47. Onboarding timeline. The system is usable from Day 1 with questionnaire-derived defaults, but requires 2–4 weeks of observation data before the full physiological model is calibrated.

20.1 Chronotype Assessment

The initial chronotype θ is derived from the Morningness-Eveningness Questionnaire (MEQ), a validated 19-item instrument. The system maps MEQ scores to the θ parameter:

Table 8. MEQ score to chronotype parameter mapping.

MEQ Score	Type	θ	Peak Window
70–86	Definite morning	−2.0	7:30–9:30am
59–69	Moderate morning	−1.0	8:30–10:30am
42–58	Intermediate	0.0	9:30–11:30am
31–41	Moderate evening	+1.0	10:30am–12:30pm
16–30	Definite evening	+2.0	11:30am–1:30pm

After 2 weeks of HRV and plan-adherence data, the system adjusts θ to match observed energy patterns.

Insight:
Self-reported chronotype often differs from physiological chronotype. The system trusts the data over the questionnaire after sufficient observations.

21 LLM Planner: The Intelligence Layer

The LLM planner operates as a refinement pass after the constraint solver, applying common-sense reasoning that formal constraints cannot capture.

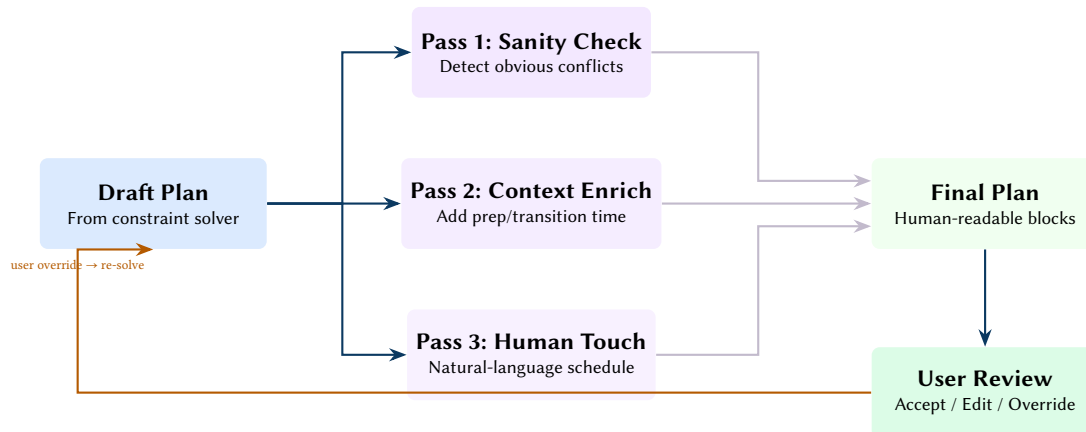


Figure 48. LLM planner three-pass refinement. The draft plan from the constraint solver is refined through sanity checking, context enrichment, and natural-language formatting before user review.

21.1 Sanity Check Examples

The LLM catches conflicts that the formal model misses:

- “Don’t schedule a phone call during the school-run window.”
- “The dentist appointment requires 30 min travel—move the preceding deep-work block earlier.”
- “You have back-to-back video calls—insert a 5-minute screen break.”
- “Tomorrow is a public holiday—remove work tasks.”

21.2 Natural Language Output

Rather than presenting a raw grid, the LLM generates a natural-language morning briefing:

Morning Briefing Example

Wednesday, March 18 — Soccer day. Your HRV is 42ms ($\alpha = 0.85$, slightly below baseline).

Morning block (8:30–10:30): PhD chapter draft. You have a 90-minute Peak window—use it for the methodology section.

Mid-morning (10:30–12:00): Email and admin. Three items flagged urgent.

Afternoon: Light creative work until 4pm, then pre-match prep. Dynamic warm-up at 5:30. Match at 7pm.

Note: Tomorrow’s plan will be adjusted for post-match recovery. Yoga is pre-scheduled for 10am Thursday.

22 Recovery Protocol Decision Tree

Different stressors require different recovery interventions. WP uses a decision tree to recommend the appropriate recovery protocol.

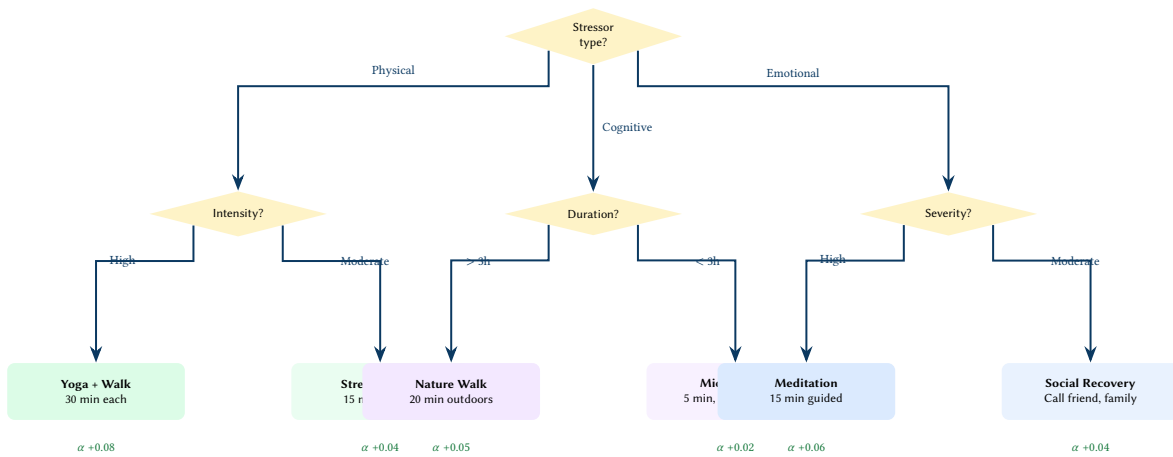


Figure 49. Recovery protocol decision tree. The type of stressor (physical, cognitive, emotional) and its intensity/duration determine the recommended recovery intervention. Each intervention has a measured impact on the recovery modifier α .

23 Analytics and Insights Dashboard

WP accumulates rich longitudinal data that powers a personal analytics dashboard. The dashboard surfaces patterns invisible to daily observation.

WeekPlanner Analytics Dashboard – Week of March 16, 2026



Figure 50. Analytics dashboard with four insight cards. Each card summarises a dimension of planning effectiveness with trend indicators and detected patterns. Sparklines show 7-day trajectories at a glance.

23.1 Insight Generation

After 30+ days of data, the system generates weekly insight reports:

Example Weekly Insight

“Your deep-work output drops 35% on days following poor sleep (< 6.5h). This happened twice this week. Consider protecting your 10pm bedtime on nights before deep-work-heavy days.”

Example Pacing Insight

“You averaged 92% energy expenditure this week (target: $\leq 80\%$). You’ve exceeded the pacing ceiling 3 of the last 4 weeks. Your HRV trend is declining. Consider dropping one commitment or adding a recovery day.”

24 Plan Lifecycle State Machine

Each daily plan progresses through a defined lifecycle with clear state transitions:

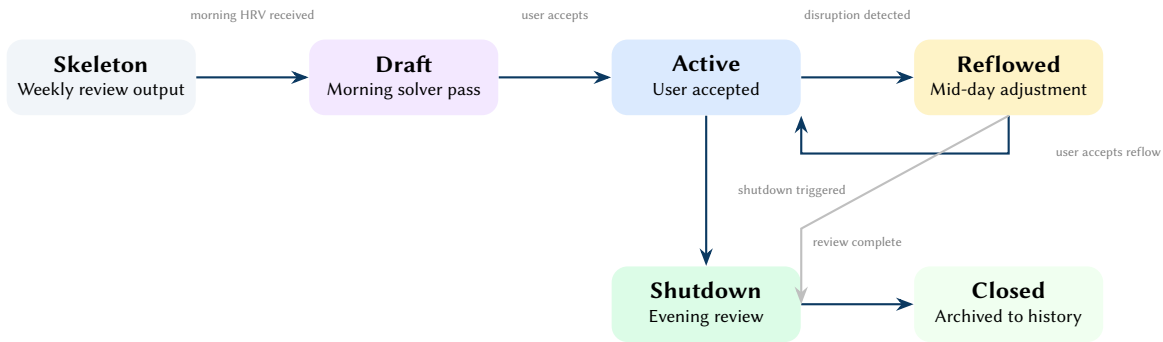


Figure 51. Plan lifecycle state machine. A plan progresses from weekly Skeleton through morning Draft, user-accepted Active, optional mid-day Reflow, evening Shutdown, and final Closed state. Each transition is triggered by a specific event.

25 Yerkes–Dodson Optimal Arousal

The current model assumes “more capacity = better scheduling”, but the Yerkes–Dodson law [12] shows that performance follows an inverted-U curve with arousal. Moderate stress actually *improves* performance on simple tasks, while complex tasks require lower arousal.

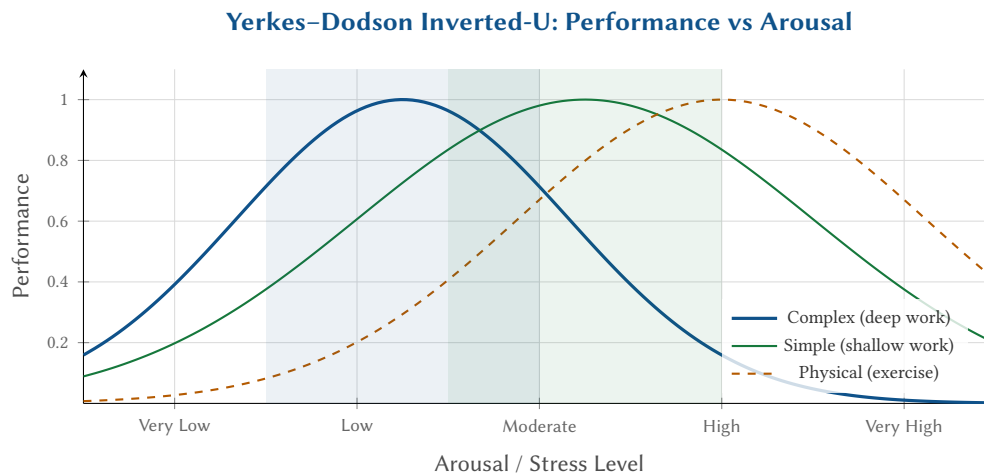
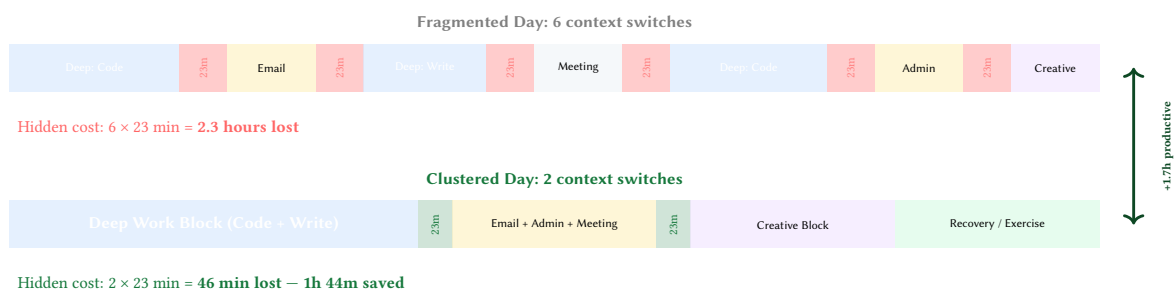


Figure 52. Yerkes–Dodson curves for three task types. Deep work peaks at low arousal; shallow work tolerates and even benefits from moderate arousal; physical tasks perform best at high arousal. The overlap zones suggest optimal scheduling: deep work when calm, exercise when energised.

26 Context-Switch Cost Visualisation

To make the cost of fragmented schedules tangible, WP can visualise the “hidden tax” of context switching:



Insight:
A future version of WP could intentionally schedule tasks at their optimal arousal level—placing complex analytical work during calm morning hours, and routine admin during the slightly elevated stress of a busy afternoon. This turns the post-lunch dip from a liability into an asset for creative work.

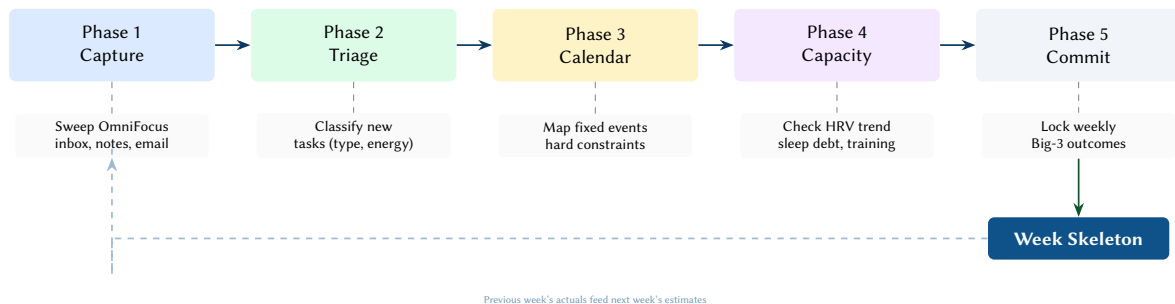
Figure 53. Context-switch cost visualisation. A fragmented day with 6 switches loses 2.3 hours to re-focus overhead (based on Mark et al.’s 23-minute average). Clustering reduces switches to 2, saving 1 hour 44 minutes of productive capacity.

27 Weekly Review Walkthrough

The weekly review is the single most important ritual in the WP system. Conducted on Sunday evening, it closes the current week's feedback loop and seeds the constraint solver with next week's fixed commitments.

27.1 Five-Phase Review Protocol

The review follows a strict five-phase protocol, each phase building on the output of the previous:



Insight:
The weekly review is where strategic intent meets tactical capacity. Without it, the system degenerates into reactive scheduling.

Figure 54. The five-phase weekly review protocol. Each phase produces artefacts consumed by the next, culminating in a Week Skeleton that seeds the daily constraint solver.

27.2 Capture Phase

The capture phase sweeps all inboxes: OmniFocus inbox, Apple Notes, email flagged items, Slack saved messages, and any paper notes from the week. The OmniSync bridge (from the existing omni project) handles the OmniFocus inbox programmatically. For each item, the system:

1. Extracts a title, optional due date, and project association.
2. Runs the LLM classifier to assign cognitive type, energy band, estimated duration, and context tags.
3. Merges duplicates using fuzzy title matching (> 0.85 cosine similarity on sentence embeddings).

27.3 Capacity Estimation

The capacity phase is unique to WP. Rather than simply asking “what do I want to do this week?”, the system first computes what the user *can* do, based on:

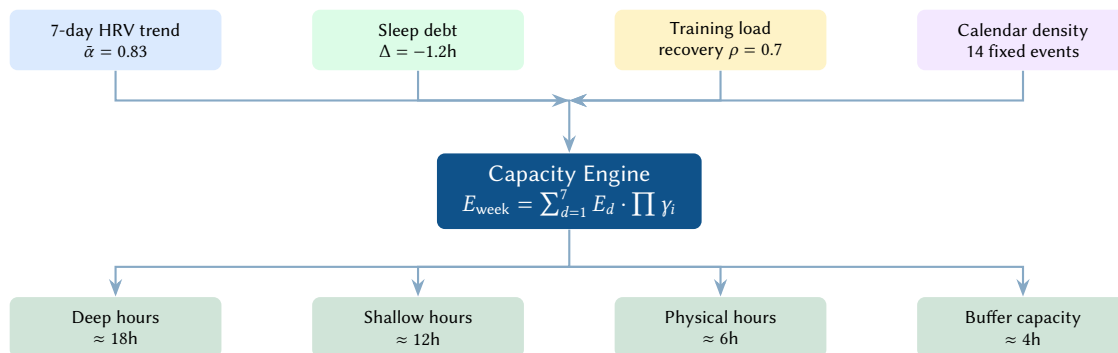


Figure 55. Weekly capacity estimation from four signal sources. The capacity engine computes available hours per energy category, accounting for physiological and calendar constraints.

The Commitment Phase then selects the week's “Big 3” outcomes — the three most important results that, if accomplished, would make the week a success. These Big 3 are locked and receive priority slot allocation before any other tasks enter the solver.

Insight:
Most planning systems ask “what do you want to do?” WP first asks “what can you do?” and then fits intent to capacity. This inversion prevents chronic overcommitment.

28 Apple Watch Real-Time Energy Monitor

While the morning HRV reading provides a daily baseline, the Apple Watch provides continuous intra-day physiological signals that can trigger plan adjustments. WP monitors three real-time streams:

28.1 Stress Index Derivation

The Stress Index (SI) is derived from the ratio of heart rate to heart-rate variability, following Shaffer & Ginsberg's framework [7]. A rising SI during a deep-work block suggests the user is either under cognitive load (expected) or experiencing external stress (unexpected). The system distinguishes these by correlating with calendar context: if the user is mid-way through a scheduled deep-work block, elevated SI is normal; if SI spikes during a recovery block, it suggests an external stressor.

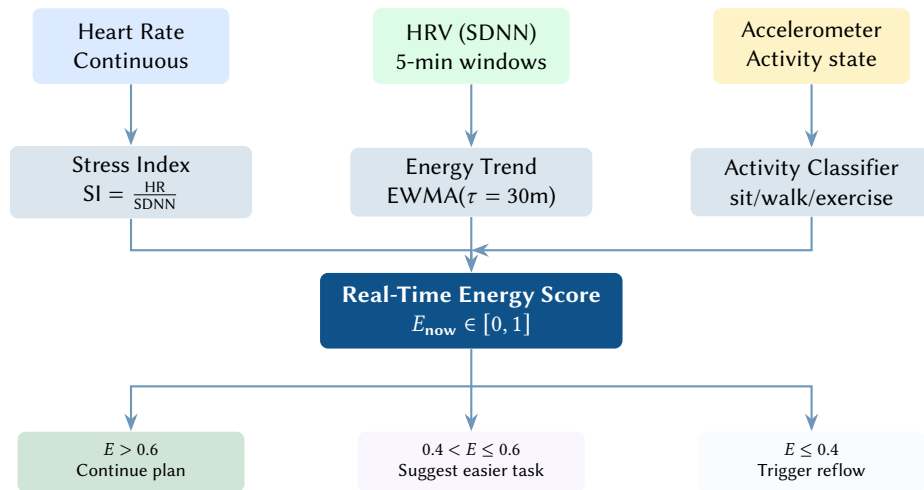


Figure 56. Apple Watch real-time energy monitoring pipeline. Three sensor streams fuse into a single energy score that gates plan continuation or triggers mid-day reflow.

Design Decision 2: Privacy-First Architecture

All physiological data processing occurs on-device. Raw sensor data never leaves the local machine. Only computed energy scores and slot-type decisions are stored, and these are held in an encrypted SQLite database alongside the task data.

28.2 Reflow Triggers

When the real-time energy score drops below threshold, the system initiates a mid-day reflow. The reflow algorithm:

1. Identifies all unstarted tasks remaining in today's plan.
2. Re-evaluates their energy requirements against the new projected capacity curve.
3. Reschedules high-energy tasks to tomorrow's Peak slot (if available) and pulls forward low-energy tasks from tomorrow or later in the week.
4. Notifies the user with a single summary: "Energy dip detected. Moved *Write proposal* to tomorrow AM. Pulled forward *Expense report* to now."

29 Notification Strategy and Alert Design

Notification design is critical: too many alerts create fatigue and undermine the system's goal of reducing cognitive overhead. Too few and the user misses important reflow events. **WP** uses a tiered notification model:

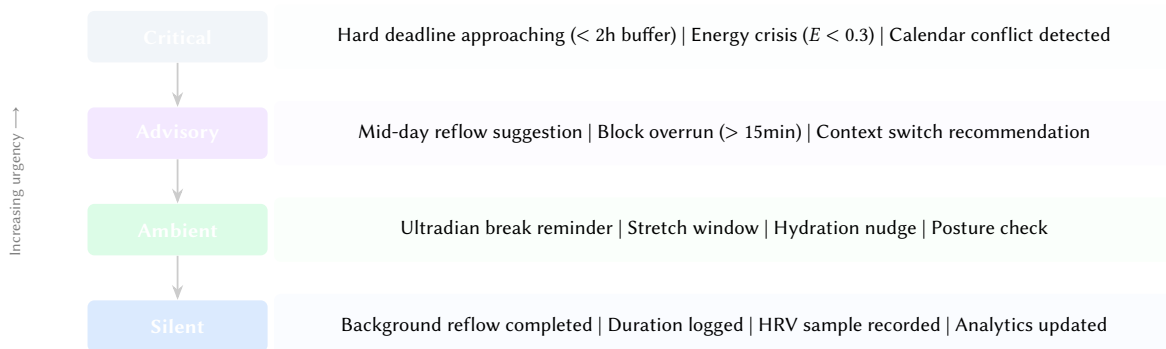


Figure 57. Four-tier notification strategy. Critical alerts interrupt immediately; advisory alerts wait for a natural break point; ambient alerts appear only in the menu bar; silent events are logged without any user-facing notification.

29.1 Focus Mode Integration

When macOS Focus Mode is active (detected via `NSWorkspace` notifications), **WP** suppresses all non-critical alerts and batches them for delivery when the focus session ends. This integrates with the Ultradian Timer: at the end of each 90-minute cycle, the system delivers any accumulated advisory notifications alongside the break prompt.

Insight:
The key insight is that advisory notifications should be deferred to natural break points. If the user is 40 minutes into a deep-work block, the reflow suggestion waits until the block ends rather than interrupting flow state.

29.2 Notification Fatigue Prevention

The system enforces a daily notification budget:

Table 9. Daily notification budget per tier.

Tier	Max/Day	Delivery	User Override
Critical	Unlimited	Immediate, sound + banner	Cannot suppress
Advisory	8	Next break point	Can defer 30 min
Ambient	12	Menu bar icon only	Can mute for day
Silent	Unlimited	Log only	N/A

30 Multi-Week Deadline Horizon

While the daily solver optimises within a single day and the weekly review sets a 7-day frame, many real-world obligations span weeks or months. **WP** maintains a rolling 4-week horizon that influences weekly capacity allocation.



Figure 58. Four-week deadline horizon with backward pressure. Deadlines in Weeks 3–4 create backward pressure that increases the allocation of preparatory work in Weeks 1–2.

30.1 Backward Pressure Algorithm

For each deadline d with estimated remaining effort E_d hours, the system computes a *backward pressure* that distributes work across available weeks:

1. Compute total deep-work capacity per week: C_w hours.
2. Subtract already-committed time to get available capacity A_w .
3. Starting from the deadline week, work backward, allocating work in $\min(A_w, E_d - \text{allocated})$ chunks.
4. If total available capacity across weeks is insufficient, flag an overcommitment warning during the weekly review.

31 Data Privacy and Security Model

WP handles deeply personal data: physiological metrics, daily schedules, task histories, and behavioural patterns. The privacy model is built on three principles:

31.1 LLM Privacy Considerations

The LLM planner layer presents a unique privacy challenge. Three deployment options are supported, in order of preference:

1. **Local model** (e.g. Llama 3 via Ollama): All inference on-device. No data leaves the machine. Limited capability but sufficient for task classification and reflow suggestions.
2. **Claude API with ephemeral mode**: Task descriptions are sent to Anthropic’s API but marked as ephemeral (not used for training). Provides superior planning intelligence.
3. **Hybrid**: Local model for classification, cloud API for weekly review planning and complex scheduling decisions.

Insight:
Backward pressure prevents the common failure mode of “I’ll start the paper next week” repeated until the deadline is imminent. The system makes the capacity gap visible weeks in advance.

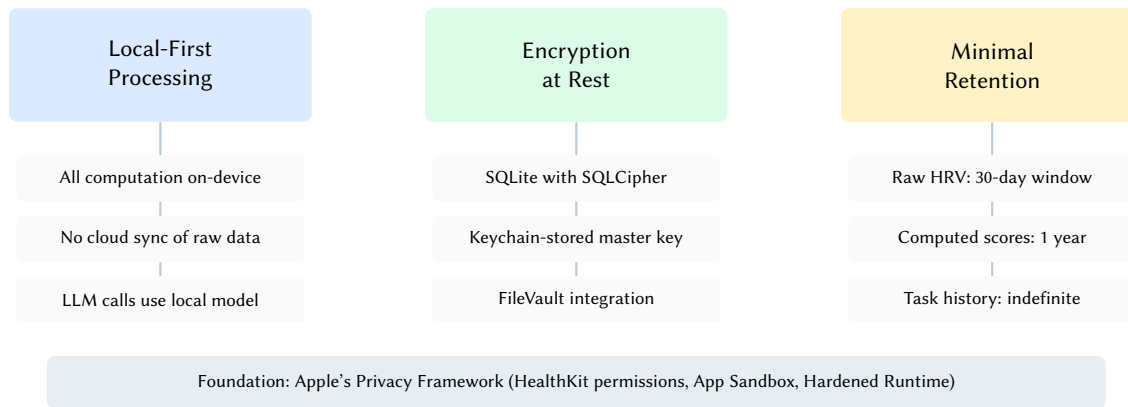


Figure 59. Three-pillar privacy model. All data processing occurs locally, storage is encrypted via SQLCipher, and retention windows are enforced to minimise data exposure.

: Data Minimisation

When calling an external LLM API, the system must strip all personally identifiable information (names, locations, specific health metrics) and transmit only abstract task descriptors: “90-min deep-analytical block at 0.8 energy” rather than “Write the Smith & Jones grant proposal”.

32 Flow State Detection and Protection

Csikszentmihalyi’s concept of flow [13] describes a state of complete absorption where productivity and satisfaction peak. WP actively detects and protects flow states using a combination of physiological and behavioural signals.

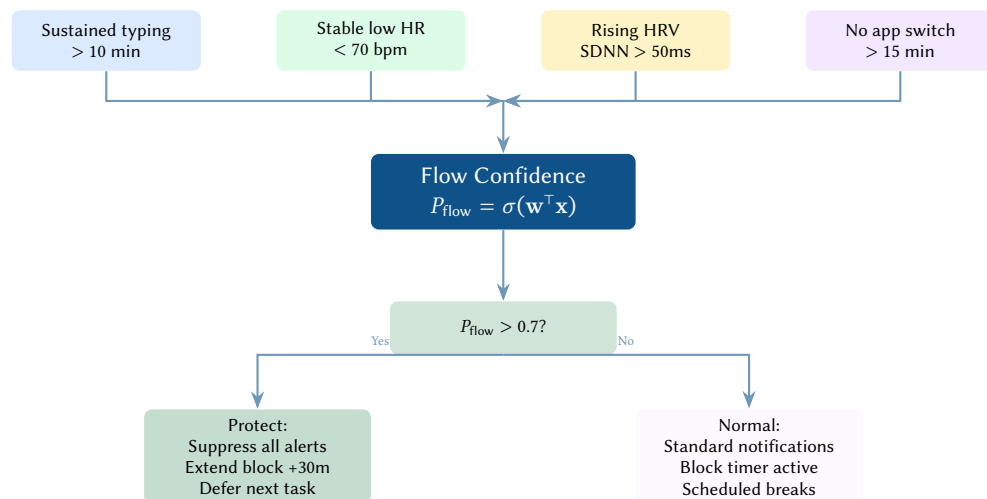


Figure 60. Flow state detection and protection. Four signals feed a logistic classifier; when flow confidence exceeds 0.7, the system enters protection mode, suppressing all interruptions and extending the current block.

When flow protection activates, the system:

- Suppresses all non-critical notifications.
- Silently extends the current time block by up to 30 minutes.
- Defers the next scheduled task (unless it has a hard deadline within 15 minutes).
- Activates macOS Focus Mode if not already active.
- Logs the flow session duration for the analytics dashboard.

Insight:
Flow state is fragile. Research shows it takes 15–25 minutes to re-enter flow after an interruption [14]. Protecting an existing flow state is worth more than any task that might replace it.

The flow detector improves over time through reinforcement: when the user manually extends a deep-work block (a strong flow signal), the model increases the weight of the signals that were active at that time.

33 Goal Alignment and Quarterly Objectives

Individual tasks gain meaning only in the context of higher-level goals [15]. WP maintains a three-tier goal hierarchy that connects daily actions to quarterly objectives:

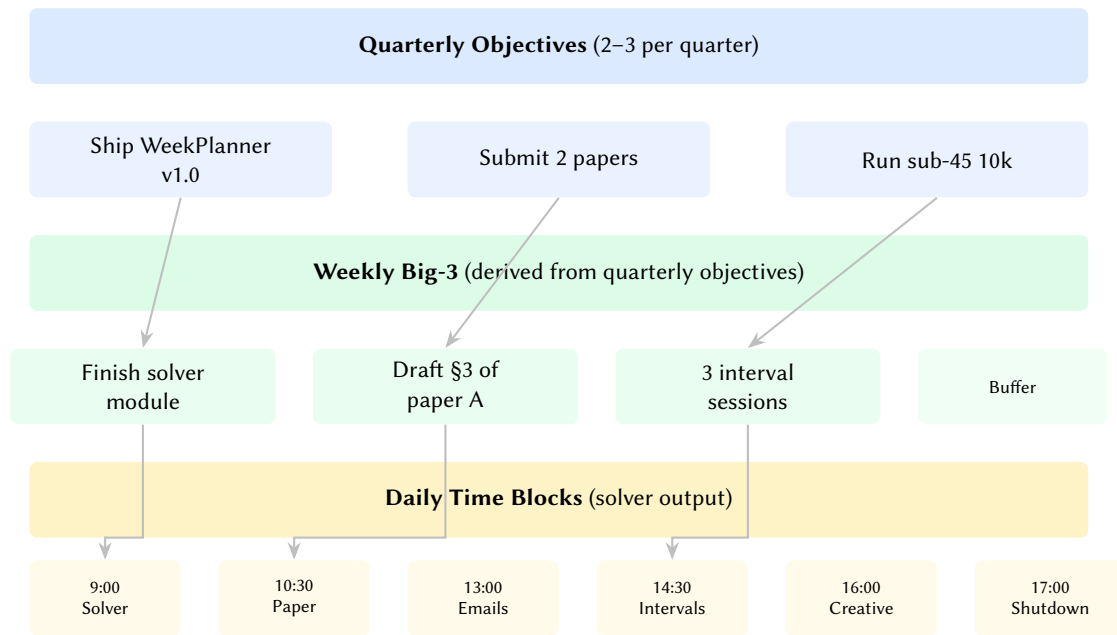


Figure 61. Three-tier goal hierarchy. Quarterly objectives decompose into weekly Big-3 outcomes, which drive daily time-block allocation. Every deep-work block should trace to a quarterly objective.

33.1 Goal Progress Tracking

Each quarterly objective has an estimated total effort (in deep-work hours) and a set of milestones. The system tracks cumulative hours invested per objective and projects completion probability:

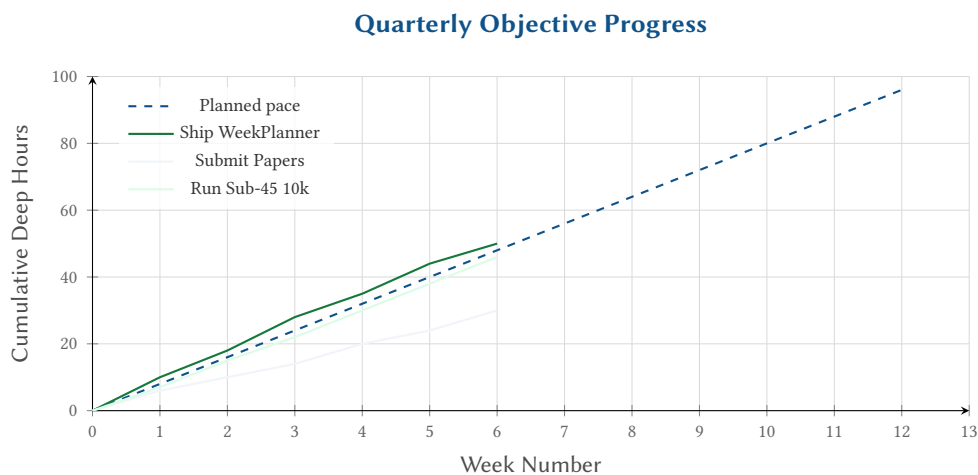


Figure 62. Goal progress burnup chart. The “Submit Papers” objective is falling behind the planned pace, triggering increased allocation in the weekly review. The 10k training is on track; WeekPlanner development is slightly ahead.

When an objective falls behind its planned pace by more than one standard deviation, the weekly review highlights it with an “attention needed” flag and suggests rebalancing the Big-3 allocation for the coming week.

34 Spaced Repetition and Learning Integration

For knowledge workers engaged in continuous learning, *WP* integrates spaced repetition scheduling based on Ebbinghaus’s forgetting curve [16]. Review sessions are treated as shallow tasks with specific timing constraints:

The scheduler places review sessions in Trough or Recovery slots, since retrieval practice requires minimal cognitive energy but benefits from slight “desirable difficulty” [17]. This means review sessions naturally fill the post-lunch dip that would otherwise be unproductive.

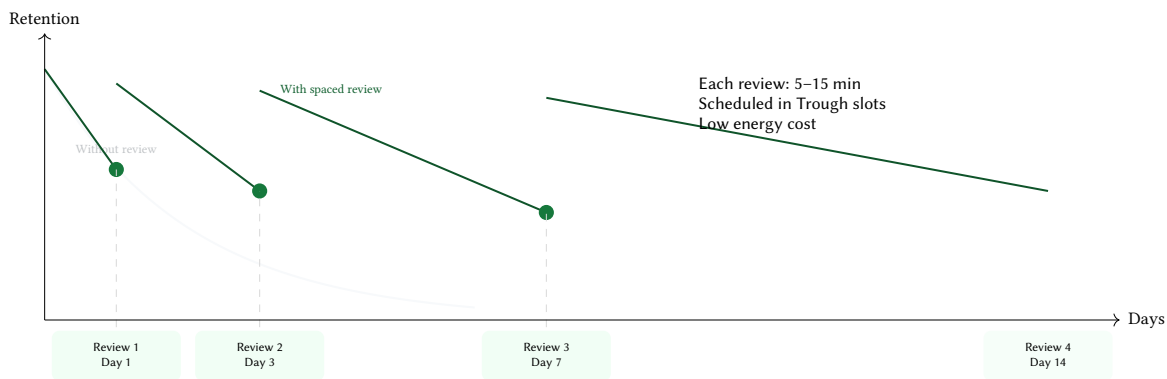


Figure 63. Spaced repetition integration. The forgetting curve (red) shows rapid decay without review. Timed reviews (green saw-tooth) maintain retention with minimal time investment, scheduled in trough slots.

35 Micro-Break Science and Scheduling

Beyond the 90-minute ultradian cycle, research demonstrates that micro-breaks (1–5 minutes) within work blocks significantly improve sustained attention [18]. **WP** schedules three types of micro-breaks:

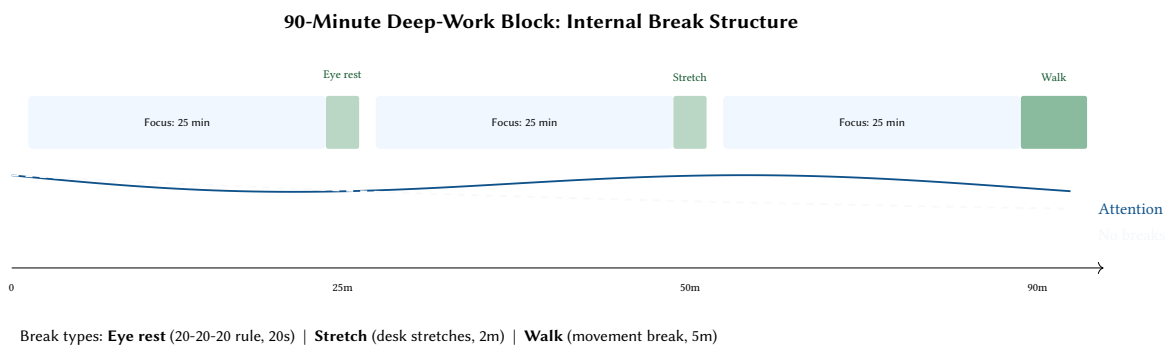
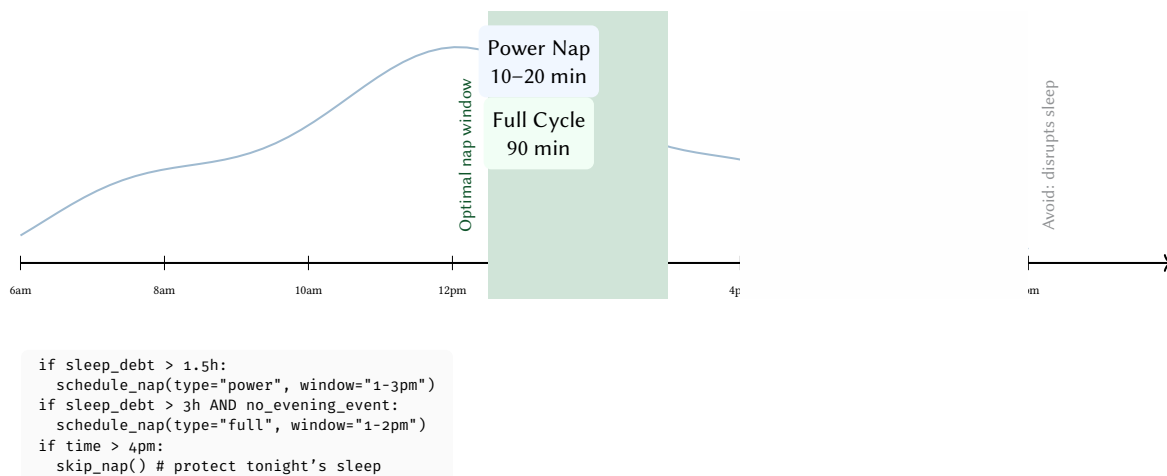


Figure 64. Internal break structure within a 90-minute deep-work block. Two micro-breaks (eye rest, stretch) maintain the attention curve (blue) above the no-breaks baseline (red dashed). The block ends with a 5-minute movement break.

36 Nap Scheduling and Power Nap Protocol

Strategic napping has been shown to restore cognitive performance [20], particularly when sleep debt is elevated. **WP** includes an optional nap scheduler that activates when $\beta(\text{TST}) < 0.8$ (i.e. the user slept less than 80% of their target).



Insight:
The 25-minute sub-blocks within the 90-minute ultradian cycle mirror the Pomodoro Technique [19], but are nested rather than independent. The user does not stop working; they simply shift gaze or stretch for 20 seconds to 2 minutes.

Figure 65. Nap scheduling decision logic. The optimal nap window (green) aligns with the post-lunch circadian dip. Naps after 4pm (red zone) are suppressed to protect night-time sleep quality.

37 Light Exposure and Circadian Entrainment

Light exposure is the primary zeitgeber (time-giver) for the circadian system [21]. **WP** incorporates light exposure reminders and tracks compliance to strengthen circadian rhythms:

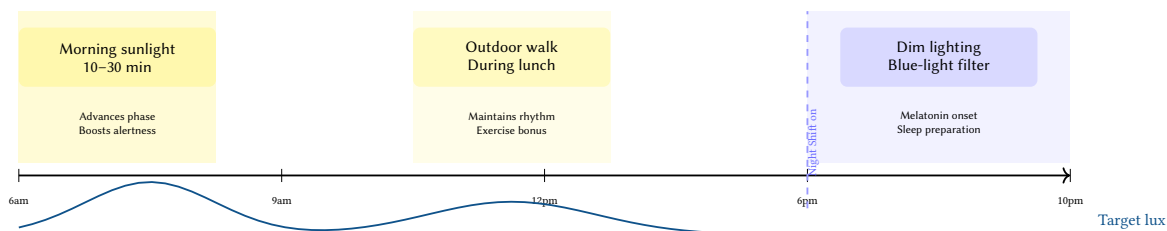


Figure 66. Light exposure protocol across the day. Morning bright light advances the circadian phase; midday outdoor exposure maintains the rhythm; evening dim lighting supports melatonin onset.

The system integrates with macOS Night Shift and schedules light exposure reminders as ambient-tier notifications:

- **Morning** (within 30 min of wake): “Step outside for 10 minutes of bright light.” Timed with the morning HRV reading.
- **Midday**: Scheduled as the lunch-break walking suggestion, combining physical movement with light exposure.
- **Evening** (bedtime – 2h): “Dimming reminder. Switch to warm lighting and enable Night Shift.”

38 Executive Function Support

Executive functions — working memory, cognitive flexibility, and inhibitory control [22] — are the neural substrate that enables effective task management. WP explicitly supports users on days when executive function is impaired (low sleep, high stress, illness):

Insight:
Adelaide’s latitude (34.9°S) means winter sunrise can be as late as 7:15am, while summer sunrise is 5:55am. The system adjusts morning light exposure timing seasonally using sunrise/sunset API data.

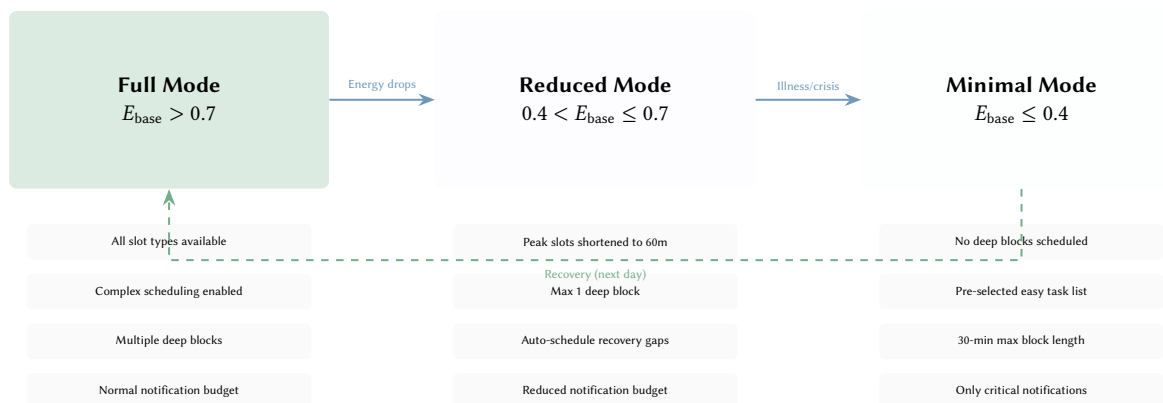


Figure 67. Three operating modes based on executive function capacity. On low-energy days, the system automatically reduces complexity, shortens blocks, and pre-selects easy tasks.

Design Decision 3: Compassionate Scheduling

When Minimal Mode activates, the system’s tone shifts from “here’s your optimised schedule” to “here are a few easy wins for today — take it gentle.” This acknowledges that not every day can be a peak performance day, and that rest *is* productive when the body demands it.

39 Retrospective Analytics Engine

Long-term improvement requires systematic reflection. The retrospective engine analyses patterns across weeks and months, surfacing insights that would be invisible in day-to-day operation.

39.1 Weekly Retrospective Report

Every Sunday, before the weekly review, the system generates a one-page retrospective covering:

1. **Plan vs. Actual:** Side-by-side comparison of planned blocks versus actual time spent, highlighting overruns and uncompleted tasks.
2. **Energy Prediction Accuracy:** How well the morning capacity estimate matched actual productivity throughout the week.
3. **Context-Switch Count:** Total interruptions and their sources, with week-over-week trend.
4. **Goal Progress:** Burnup chart progress for each quarterly objective.
5. **Physiological Trends:** HRV trend, sleep debt trajectory, and recovery compliance.

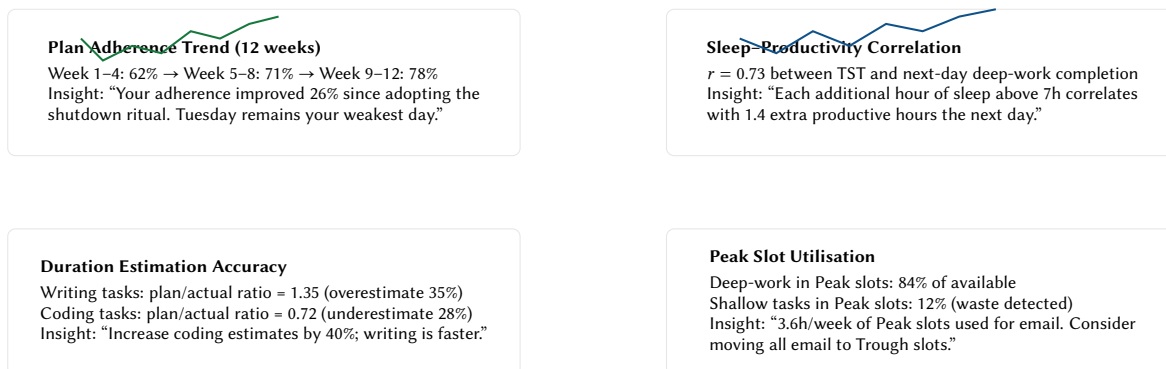


Figure 68. Retrospective insight cards generated from 12 weeks of data. Each card surfaces an actionable pattern with a specific recommendation.

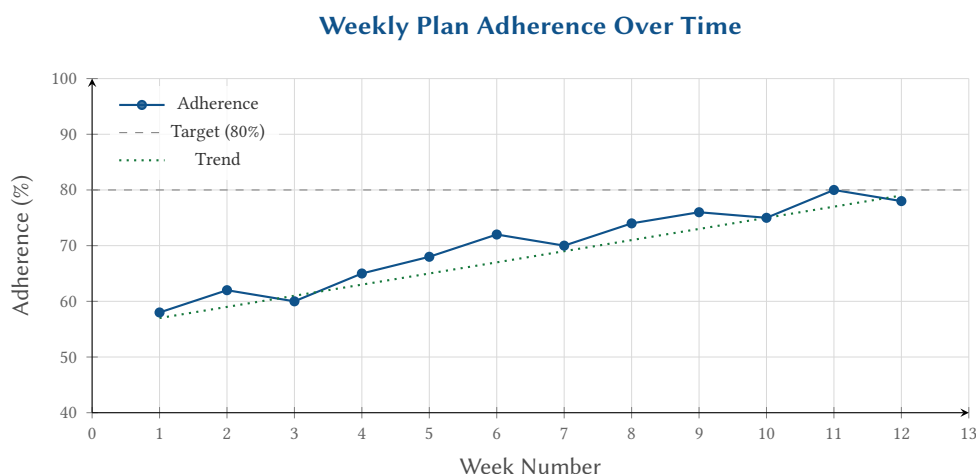


Figure 69. Plan adherence trending upward over 12 weeks, approaching the 80% target. The positive trend validates that the system’s feedback loops are driving continuous improvement.

40 API Design for Third-Party Integration

While **WP** is designed as a local-first system, a lightweight API enables integration with external tools. The API is exposed via a local Unix domain socket (no network exposure by default).

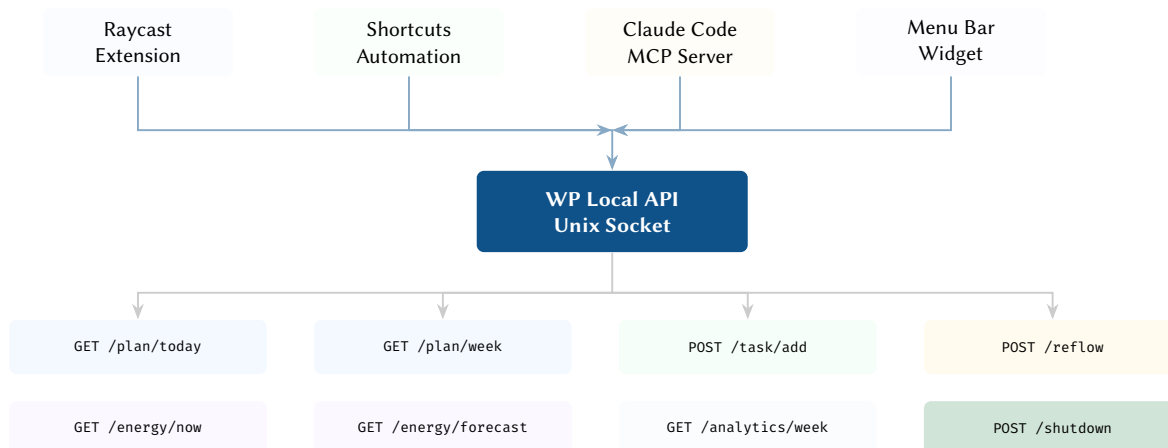


Figure 70. Local API architecture. Four client types connect via Unix domain socket to eight core endpoints. The MCP Server integration enables Claude Code to query and modify the plan conversationally.

40.1 MCP Server Integration

The most powerful integration is the **WP** MCP (Model Context Protocol) server, which exposes the planning system to Claude Code. This enables conversational plan management:

- “What’s my plan for today?” — queries `GET /plan/today`.
- “I finished the code review early, what should I do next?” — triggers `POST /reflow` with updated actuals.

- “Add a 2-hour deep block for grant writing tomorrow morning” — calls POST /task/add with constraints.
- “How’s my energy looking for the afternoon?” — queries GET /energy/forecast.

This closes the loop between the AI assistant and the planning system, creating a unified interface where the user can manage their day through natural language.

41 Constraint Solver: Detailed Pseudocode and Worked Example

This section provides a complete algorithmic specification of the greedy assignment loop, followed by a worked example that traces the solver through a concrete day, showing every decision, constraint check, and relaxation step in detail.

41.1 Full Algorithm

The solver operates in six passes over a sorted task list. Pass order encodes priority: fixed calendar events are immovable anchors; deadline tasks are placed before willpower is depleted; deep work targets peaks; physical activity targets sustain zones; shallow tasks fill gaps; recovery fills the remainder.

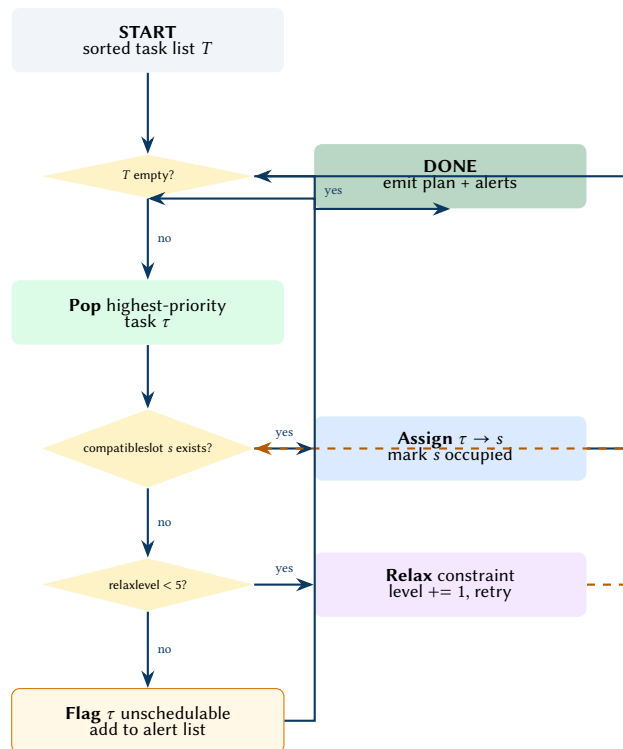


Figure 71. Greedy solver control flow. Each task is popped from the priority-sorted list and placed into the best compatible slot. If no compatible slot exists the solver progressively relaxes constraints up to level 5 before flagging the task as unschedulable.

41.2 Worked Example: Six Tasks, One Day

Consider a Monday with the following six energy slots and six candidate tasks. We trace every assignment decision including one relaxation.

Insight:
Even a single relaxation step should be surfaced to the user. The log entry for T6 reads: “Deep Research placed in Sustain (Evening) because both Peak slots were occupied. Consider deferring T2 to free a Peak slot if this task is higher value.” This transparency is what distinguishes WP from opaque auto-schedulers.


```

Algorithm GreedySolver(tasks T, slots S, relax_cap=5):

    sort T by priority_score DESC

    for  $\tau$  in T:

        placed = False

        for level in  $\emptyset$ ..relax_cap:

            s = best_compatible_slot( $\tau$ , S, relax=level)

            if s is not None:

                assign( $\tau$ , s); mark_occupied(s)

                log_relaxation( $\tau$ , level) if level > 0

                placed = True; break

            if not placed:

                flag_unschedulable( $\tau$ )

    return emit_plan(S), alerts

```

Figure 72. Pseudocode for the greedy solver. `best_compatible_slot` returns the highest-capacity compatible slot at the given relaxation level, or `None` if no slot qualifies.

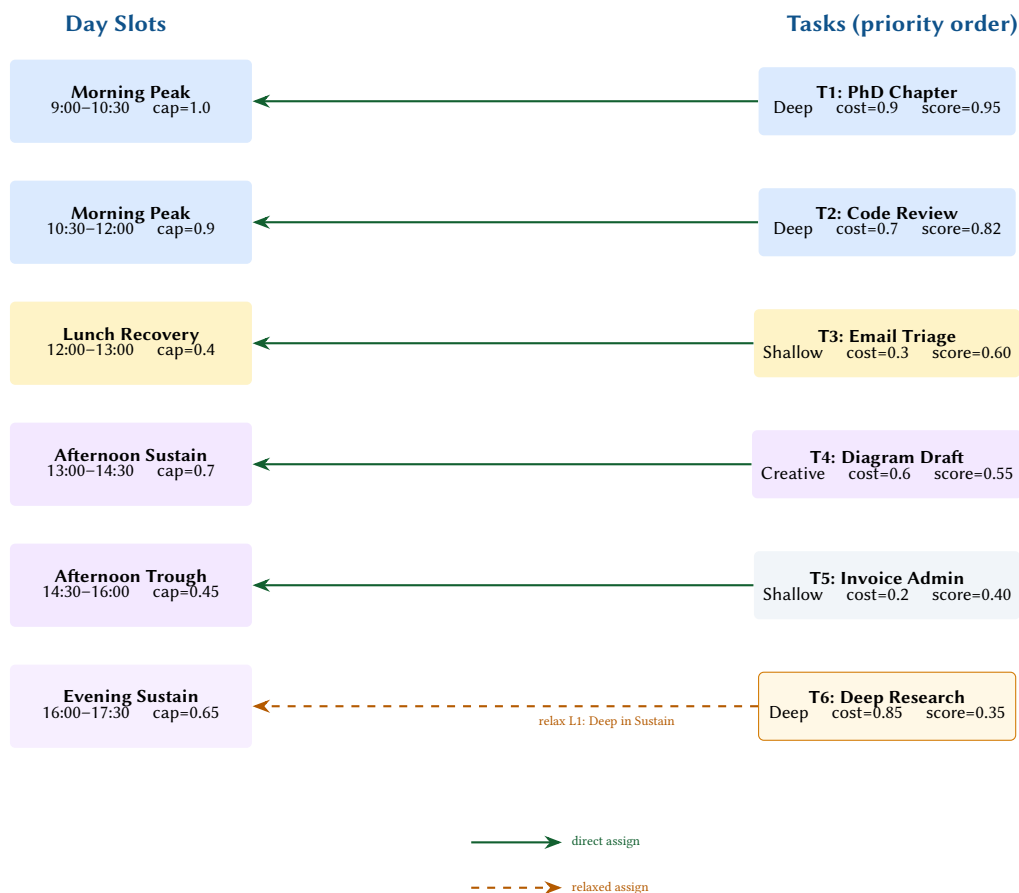


Figure 73. Worked example: six tasks assigned to six day slots. Tasks T1–T5 are placed without relaxation. T6 (Deep Research) cannot reach a Peak slot—all are occupied—so the solver applies relaxation level 1 (Deep allowed in Sustain) and places it in the Evening Sustain slot with a logged warning.

42 Migration Path: From OmniFocus to WeekPlanner

Adopting a new planning system carries risk proportional to reliance on the existing system. A three-phase migration strategy de-risks the transition by maintaining OmniFocus as the source-of-truth throughout Phases 1 and 2, only ceding control in Phase 3 when user confidence is established.

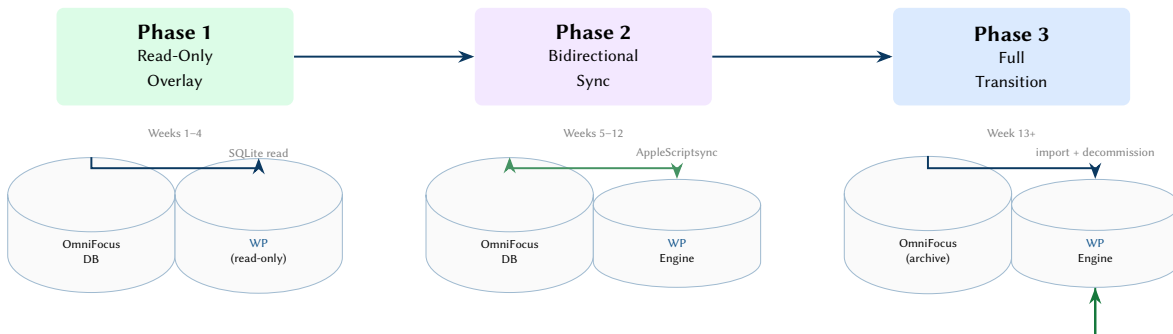


Figure 74. Three-phase migration from OmniFocus to WP. Phase 1 is purely additive—WP reads OmniFocus data without modification. Phase 2 introduces bidirectional sync. Phase 3 completes the transition.

42.1 Concept Mapping

The following table maps OmniFocus constructs to their WP equivalents, guiding migration of existing workflow conventions.

Table 10. OmniFocus to WP concept mapping.

OmniFocus Concept	WP Equivalent	Notes
Perspective	View / Filter	User-defined task subsets
Context (legacy)	Area	Location/tool grouping
Flag	Energy Band priority	Flags map to Deep or high-urgency
Forecast	Day Skeleton	Merged calendar + task view
Review	Weekly Review protocol	Structured 5-phase process
Defer date	Not-before constraint	Prevents early scheduling
Due date	Hard deadline	Triggers urgency sigmoid
Project	Project (preserved)	Direct mapping
Tags	Tags + Energy type	Extended with energy semantics
Sequential action	Dependency edge	Directed Acyclic Graph (DAG) constraint
Parallel action	Unconstrained tasks	Freely assignable
On Hold project	Parked project	Excluded from scheduling

42.2 Risk Mitigation per Phase

Design Decision 4: Phase 1 Risk: Zero write-back contamination

WP in Phase 1 opens the OmniFocus SQLite database in `PRAGMA journal_mode=WAL; PRAGMA query_only=ON; mode.` No writes are possible. A file-system level read-only mount provides a second safety layer. Risk: *low*.

Design Decision 5: Phase 2 Risk: Sync conflicts

Bidirectional sync uses OmniFocus AppleScript `complete task` and `add task` verbs. Conflict resolution policy: OmniFocus wins on completion state; WP wins on scheduling metadata (energy type, estimated duration). All sync operations are journalled for rollback. Risk: *medium*. Mitigation: nightly OmniFocus backup before any sync run.

Design Decision 6: Phase 3 Risk: Data loss on decommission

Before decommissioning OmniFocus, a full export of all projects and tasks to both OmniFocus XML and WP native format is verified by record-count checksums. OmniFocus licence remains active for 90 days post-transition as emergency fallback. Risk: *low* with procedure.

43 Accessibility and Inclusive Design

WP targets WCAG 2.1 Level AA compliance for all user-facing surfaces [23]. Accessibility is treated as a first-class design constraint, not a post-hoc audit [24].

43.1 Colour-Blind Safe Palette

The five layer colours and two accent colours are selected from a colour-blind safe palette verified against Deuteranopia, Protanopia, and Tritanopia simulation. The palette uses the Paul Tol “Muted” scheme.

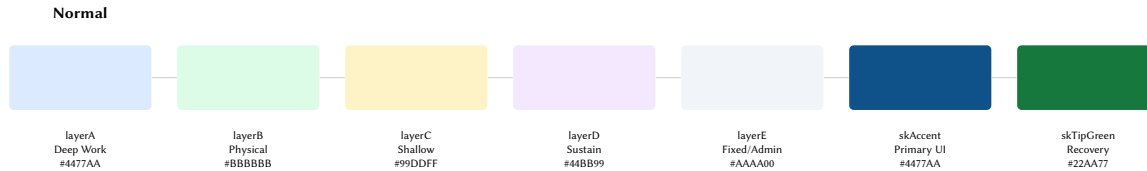


Figure 75. Colour-blind safe palette used throughout WP. All seven colours are distinguishable under Deuteranopia, Protanopia, and Tritanopia simulation. Each colour is paired with a distinct shape or icon in the UI to satisfy the WCAG 1.4.1 non-colour redundancy requirement.

43.2 WCAG 2.1 Compliance Targets

Table 11. WCAG 2.1 compliance targets for WP surfaces.

Criterion	Level	Surface	Status
1.4.1 Use of Colour	A	All	Colour + shape redundancy
1.4.3 Contrast Ratio (text)	AA	Menu bar, TUI	$\geq 4.5 : 1$
1.4.11 Non-text Contrast	AA	Charts, icons	$\geq 3 : 1$
2.1.1 Keyboard	A	All	Full keyboard nav
2.3.1 Three Flashes	A	Animations	No flashing content
4.1.2 Name, Role, Value	A	macOS UI	VoiceOver labels

43.3 Screen Reader and Reduced-Motion Support

The macOS menu bar app uses NSAccessibility labels on all custom views. The slot grid exposes each block as an accessible element with a label of the form: “Morning Peak, 9:00–10:30, PhD Chapter, Deep Work, 90 minutes.” Time-block animations honour the system `NSWorkspaceAccessibilityDisplayOptionsDidChange` notification: when Reduce Motion is enabled, all transitions become instant cuts.

43.4 Cognitive Accessibility: Simplified Mode

: Simplified Mode

When enabled, Simplified Mode reduces the daily view to three slots (Morning, Afternoon, Evening) and presents at most three tasks per slot. The energy score system is hidden; tasks are presented as a plain ordered list. This mode is intended for high-cognitive-load days or for users new to the system.

44 Failure Modes and Graceful Degradation

A physiologically-informed planner that fails silently is worse than no planner at all: the user may act on stale or missing data without realising it. WP categorises every external dependency by failure mode, detection method, fallback behaviour, and recovery path.

44.1 Degradation Levels

44.2 Failure Decision Tree

Insight:
The most important failure to handle gracefully is “Watch not worn last night.” This is the most common real-world scenario: the system must fall back to the 7-day rolling HRV average without alerting the user every morning. The alert is reserved for ≥ 3 consecutive missed readings.

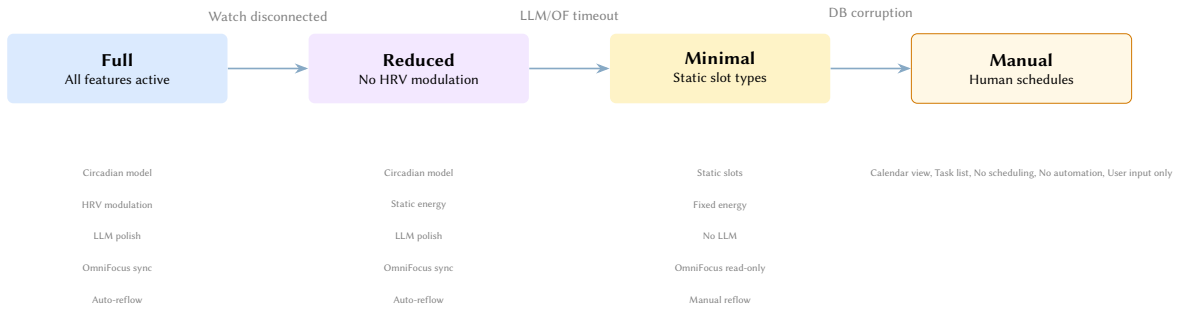


Figure 76. Degradation levels and feature availability. Each transition is triggered by a specific class of dependency failure. The system always remains usable—it never presents a blank screen or unhandled exception.

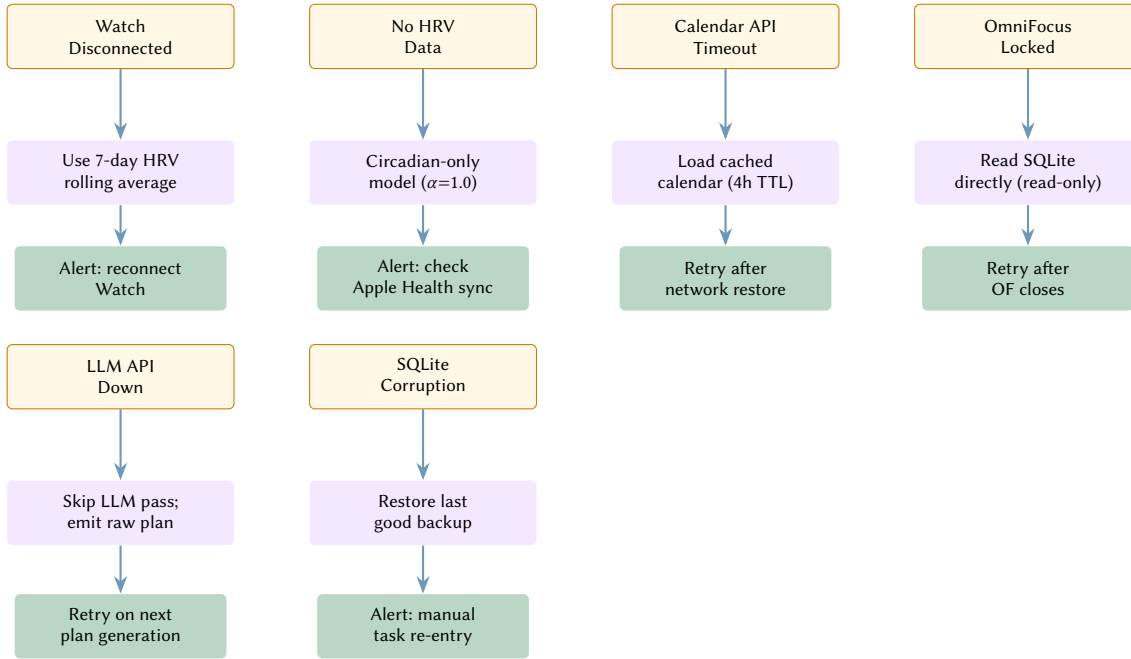


Figure 77. Failure decision tree. Each failure mode (red) has a defined fallback action (blue) and a recovery path (green). No failure results in an unhandled exception or missing plan.

45 User Testing Methodology

WP follows an N=1 self-experimentation protocol. While this limits external validity, it is appropriate for a tool built around one user's physiology and workflow [2]. The protocol is designed to yield actionable calibration data within 12 weeks.

45.1 Study Design

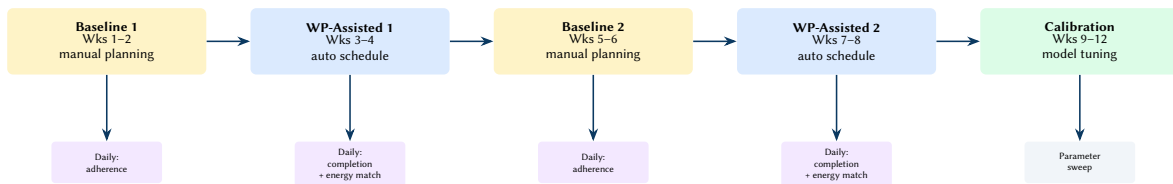


Figure 78. 12-week A/B protocol. Baseline periods use manual planning with identical task capture; WP-Assisted periods engage the full scheduling engine. The final calibration period tunes model parameters using accumulated data.

45.2 Metrics

45.3 Statistical Analysis Plan

The primary analysis is a paired t -test on plan adherence, comparing the two Baseline weeks against the two WP-Assisted weeks. Effect size is reported as Cohen's d . Secondary analyses use a within-subjects one-way ANOVA on Energy Match Score across four 2-week periods, with Tukey HSD post-hoc correction. All tests are two-tailed at $\alpha = 0.05$.

Table 12. Primary and secondary metrics for the self-experimentation protocol.

Metric	Definition	Analysis
Plan Adherence	% tasks completed as scheduled	Paired <i>t</i> -test (B vs WP)
Task Completion Rate	% of planned tasks completed/day	Paired <i>t</i> -test
Energy Match Score	Self-rated slot–task energy fit (1–5)	Within-subjects ANOVA
Duration Accuracy	estimated – actual / estimated	Paired <i>t</i> -test
Overcommitment Rate	% weeks exceeding capacity alert	Descriptive
Reflow Frequency	Mid-day reflows triggered / week	Descriptive

: N=1 Validity Boundary

The study design is explicitly not intended to generalise to other users. Its purpose is to calibrate model parameters (θ , λ , willpower drain, HRV correction factors) to the individual. Findings that replicate across all four periods will inform the default parameters shipped with WP.

46 Full Weekly Review Worked Example

This section walks through a complete Sunday evening weekly review using realistic sample data, tracing each of the five review phases from raw inputs to a finished Week Skeleton.

46.1 Input State: Sunday Evening

OmniFocus Inbox (10 items awaiting triage):

1. Write introduction section for Chapter 4 (*Deep, est. 90 min*)
2. Book dentist appointment (*Shallow, est. 5 min*)
3. Review grant budget spreadsheet (*Deep, est. 45 min*)
4. Reply to supervisor email re: thesis timeline (*Shallow, est. 15 min*)
5. Prepare cycling route for Saturday group ride (*Shallow, est. 10 min*)
6. Debug HRV parser race condition (*Deep, est. 2 h*)
7. Update WeekPlanner bib file with new citations (*Shallow, est. 20 min*)
8. Read “Algorithms for Scheduling” paper (*Deep, est. 60 min*)
9. Call insurance about claim (*Shallow, est. 20 min*)
10. Sketch Week 13 diagram layout (*Creative, est. 30 min*)

Calendar for next week: Monday 09:00 Research group meeting (90 min); Tuesday 14:00 Dentist (60 min); Thursday 17:30 Soccer training (90 min); Saturday 08:00 Group ride (3 h).

Physiological indicators: HRV trend = declining (3-day EWMA: 58 → 54 → 49 ms, baseline 62 ms); Sleep debt = 1.5 h accumulated over 5 days; $\alpha_{\text{HRV}} = 0.82$.

46.2 Phase-by-Phase Walkthrough

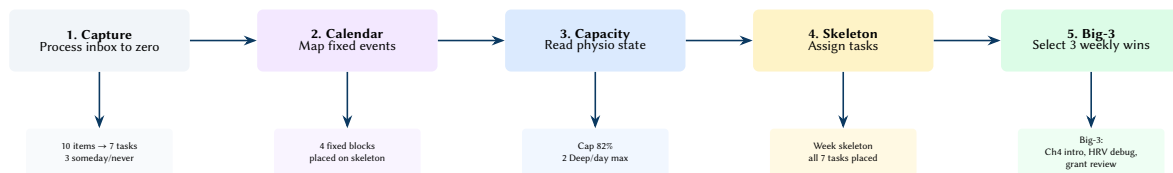


Figure 79. Five review phases with outputs for the worked example. The declining HRV and sleep debt reduce capacity to 82%, capping the week at two Deep tasks per day instead of the nominal three.

Phase 1 — Capture: The 10 inbox items are processed. Items 2 (book dentist) and 9 (insurance call) are deferred to Tuesday afternoon as they require phone calls during business hours. Item 5 (cycling route) becomes a Saturday morning task automatically from calendar context. Items 6 (HRV debug) and 1 (Chapter 4 intro) are flagged as Big-3 candidates. Items 3, 8, 10 are active tasks. Item 7 is trivial (<5 min) and done immediately. Total active tasks for the week: 7.

Phase 2 — Calendar: Four fixed blocks are placed: Monday meeting, Tuesday dentist, Thursday soccer, Saturday ride. These consume Monday 09:00–10:30, Tuesday 14:00–15:00, Thursday 17:30–19:00, and Saturday 08:00–11:00. Deep work windows are protected: Monday post-meeting (10:30–12:00), Tuesday morning (09:00–12:00), Wednesday morning, Friday morning.

Phase 3 — Capacity: HRV at 79% of baseline ($49/62 = 0.79$), sleep debt modifier $\beta = 0.95$ (1.5 h debt). Combined $\alpha \cdot \beta = 0.75$. The system recommends capping Deep work at 2 sessions/day (vs. nominal 3) and scheduling a 30-minute passive recovery block each afternoon.

Phase 4 — Skeleton: Tasks are placed by the solver. HRV debug (Deep, 2 h) takes Monday 10:30–12:30. Chapter 4 intro (Deep, 90 min) takes Tuesday 09:00–10:30. Grant review (Deep, 45 min) takes Wednesday 09:00–09:45. Remaining shallow tasks fill afternoon Sustain slots.

Phase 5 — Big-3: HRV debug, Chapter 4 intro, Grant review. Rationale: all three are deadline-proximate (thesis timeline pressure), require Deep energy, and represent concrete measurable progress.

46.3 Resulting Week Skeleton

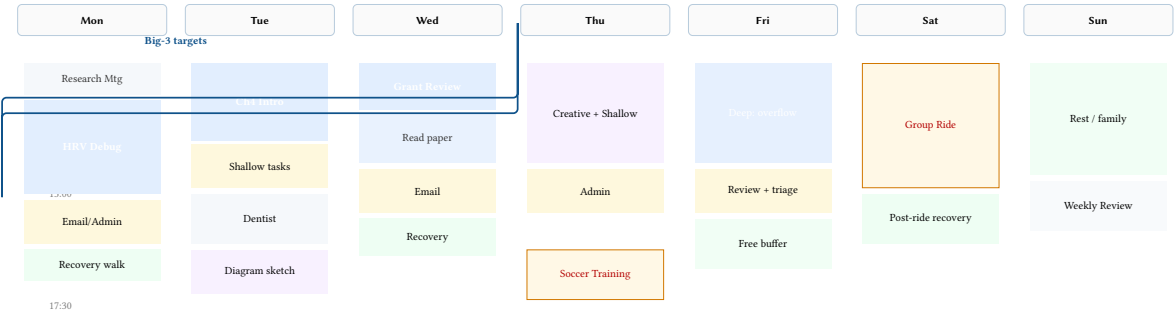


Figure 80. Week Skeleton produced by the Sunday review. Big-3 tasks (HRV Debug, Ch4 Intro, Grant Review) are protected in Monday/Tuesday/Wednesday morning Peak slots. Declining HRV reduces Deep sessions to two per day and adds afternoon recovery blocks throughout.

47 Desktop Application: Interface Design

The desktop application (wp_gui) provides eight visualisation views for analysing task data, exploring circadian capacity, and deep-diving into OmniFocus project/tag structure. Each view is rendered with egui and follows the colour-blind safe palette used throughout this report. Figures 81–88 show wireframe mockups of the planned interface.

Insight:
The declining HRV and accumulated sleep debt are visible in the resulting skeleton: the plan is deliberately lighter than a “good week” plan would be. The system’s capacity-awareness prevents the common failure of planning a heroic week on a depleted body.

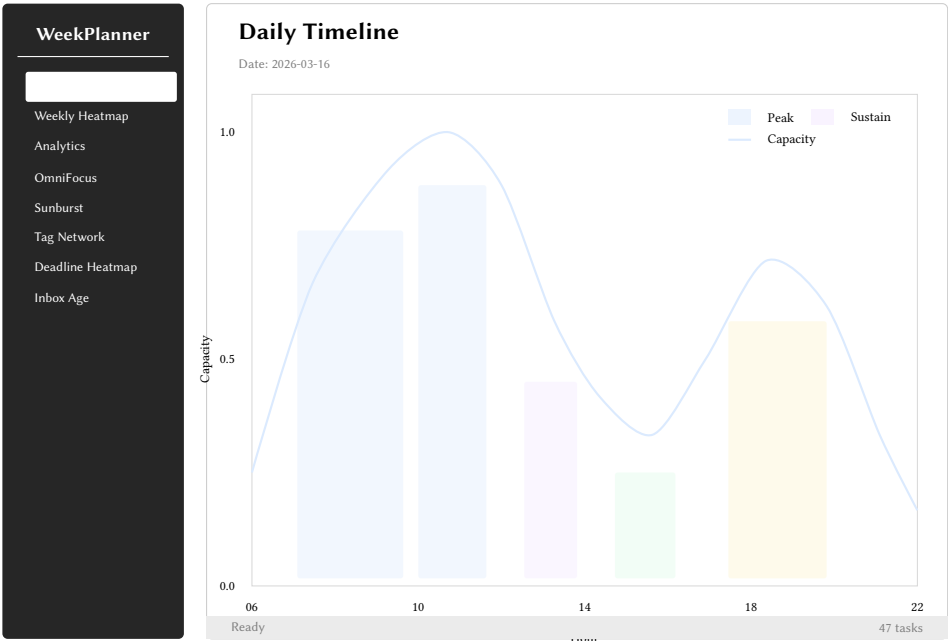


Figure 81. Daily Timeline view: the circadian capacity curve (solid line) overlaid with time-slot rectangles coloured by slot type. The vertical extent of each bar represents the slot’s capacity fraction.

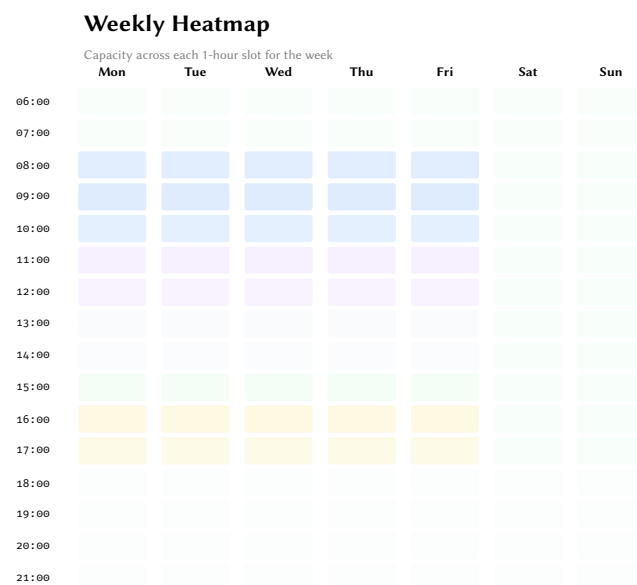


Figure 82. Weekly Heatmap view: a 7×16 grid of one-hour slots. Cell opacity encodes circadian capacity; colour encodes slot type (blue = Peak, teal = Sustain, yellow = Trough, grey = Recovery, cyan = Creative).

Analytics Dashboard

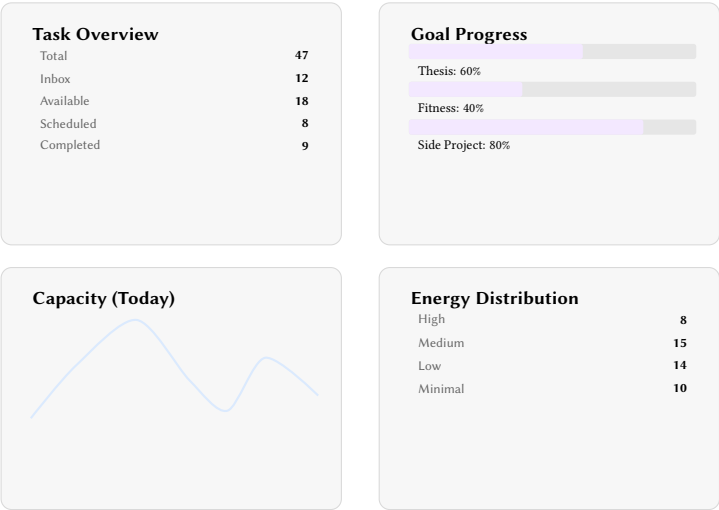


Figure 83. Analytics Dashboard: four cards showing task status counts, goal progress bars, a circadian capacity sparkline, and energy-band distribution.

OmniFocus Overview

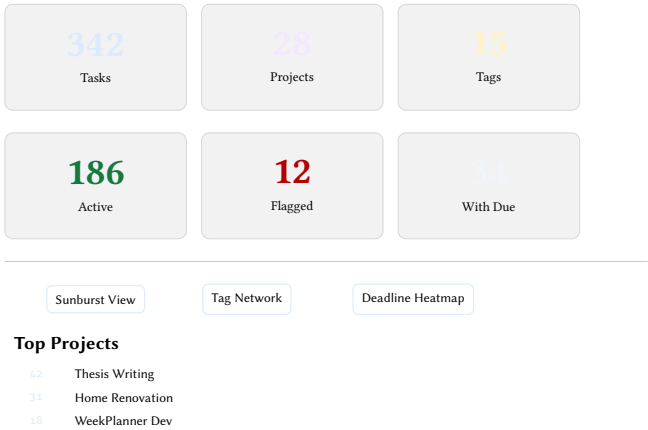


Figure 84. OmniFocus Overview: summary statistics, sub-navigation to specialised views, and a ranked project list.

Project Sunburst

Folder > Project > Task hierarchy from OmniFocus

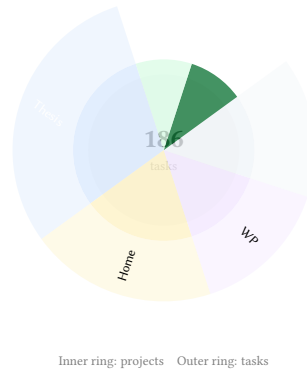


Figure 85. Project Sunburst: nested rings showing the folder–project–task hierarchy. Arc width is proportional to task count; colour distinguishes projects.

Tag Co-occurrence Network

Tags connected when they appear on the same task

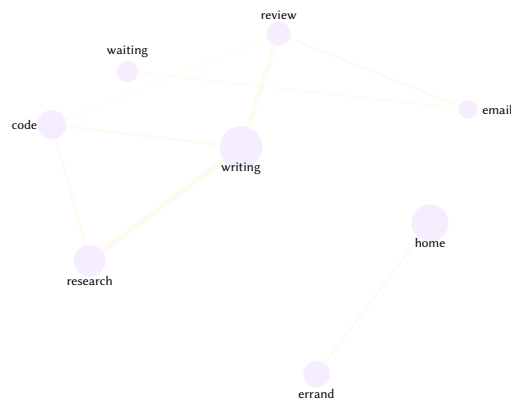


Figure 86. Tag Co-occurrence Network: a force-directed graph where nodes represent tags (sized by usage count) and edges connect tags that co-occur on the same task (thickness = co-occurrence count).

Deadline Heatmap

Task due-date density by calendar day

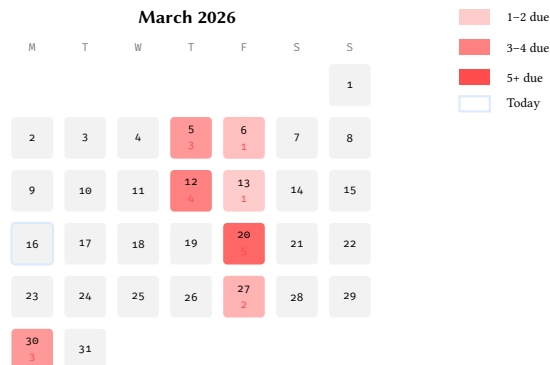


Figure 87. Deadline Heatmap: a calendar month grid where cell colour intensity encodes the number of tasks due on each day. Today's cell is outlined in the accent colour.

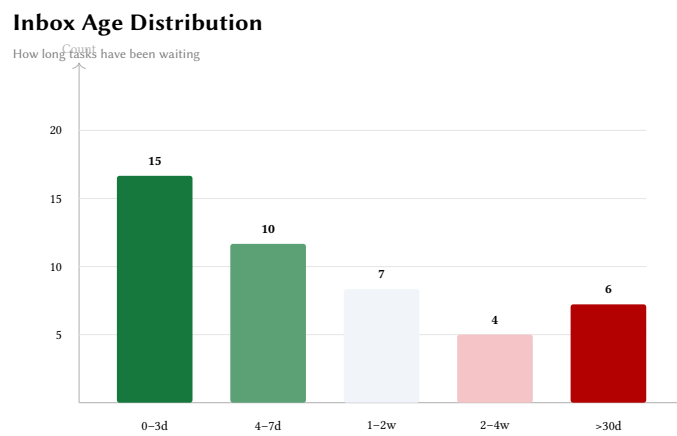


Figure 88. Inbox Age Distribution: a histogram of task ages using a green-to-red colour ramp. Fresh tasks (0–3 days) appear in green; stale tasks (>30 days) in red, prompting review or archival.

48 Open Questions and Future Work

Several design decisions remain open:

1. **Constraint solver choice.** A simple greedy allocator may suffice for Phase 1; a proper CSP solver (e.g. Google OR-Tools) provides optimality guarantees but adds dependency weight.
2. **Calendar write-back.** Should WP write time blocks back to Apple Calendar as events, creating a visual daily plan? This is high-value but requires careful conflict handling.
3. **Notification strategy.** Mid-day reflow could push notifications (“your 2pm deep-work block should shift to 3pm given your meeting overrun”), but notification fatigue is a real risk.
4. **Multi-week horizon.** The current model plans one week at a time. Extending to a rolling 2–4 week view would better support deadline-driven work and training periodisation.
5. **Social calendar awareness.** Family commitments, school pickups, and partner schedules create hard constraints that the system must respect but currently has no source for beyond calendar entries.
6. **Yerkes–Dodson integration.** The current model assumes “more capacity = better”, but moderate stress actually improves performance [12]. A future version could intentionally schedule mildly challenging work during moderate-energy periods.
7. **Training periodisation.** For users with structured training plans (running programmes, strength cycles), WP could import the training calendar and automatically derive taper/recovery constraints from the prescribed intensity.
8. **Seasonal variation.** Daylight hours affect circadian rhythms, particularly in Adelaide’s latitude. Winter mornings may shift the effective Peak window later, requiring seasonal calibration of ϕ_0 .

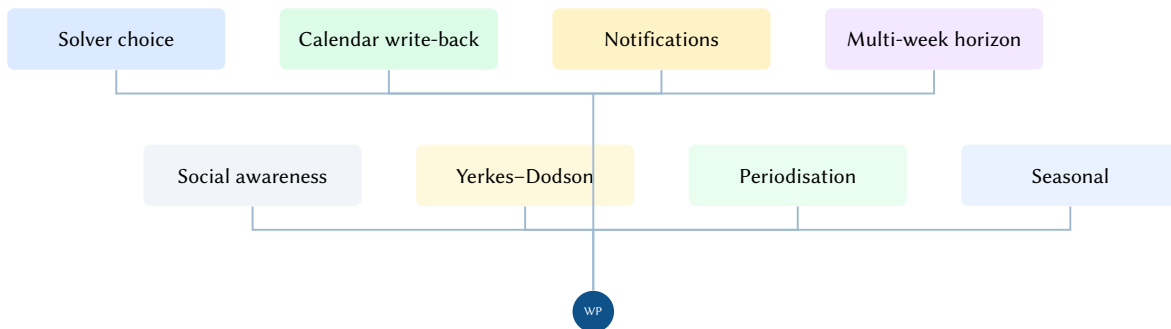


Figure 89. Open design questions radiating from the WP core. Each represents a future-work axis that could extend the system’s capabilities.

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A Slot Type Decision Table

The following table provides the complete decision logic for mapping circadian capacity values to slot types:

Table 13. Slot type thresholds (normalised capacity $C \in [0, 1]$).

Capacity Range	Slot Type	Permitted Tasks	Max Energy/Slot
$C \geq 0.75$	Peak	Deep, Shallow, Physical, Recovery	0.9
$0.50 \leq C < 0.75$	Sustain	Shallow, Physical, Recovery	0.6
$0.30 \leq C < 0.50$	Trough	Shallow, Recovery	0.3
$C < 0.30$	Recovery	Recovery only	0.1

B Weekly Constraint Propagation Rules

Table 14. Complete weekly constraint propagation rules.

Trigger	Day -1	Day +1	Day +2	Mandatory Blocks
Soccer match	Taper 40%	$\rho = 0.65$	$\rho = 0.85$	Stretch (pre+post), Yoga (D+1)
Gym session	—	$\rho = 0.75$	—	Cool-down (post)
Long run	Light day	$\rho = 0.80$	—	Stretch (post)
2 consecutive high days	—	Force light	—	30-min walk minimum
TST < 7h $\times 2$	—	$\beta = 0.72$	$\beta = 0.85$	Extra recovery block

Revision History

Version	Date	Changes
0.2.0	March 2026	Expanded diagrams; Recovery science; Data model; Solver detail
0.1.0	March 2026	Initial Architecture and Constraint Model